2026 AI 創新獎

2026 AI Innovation Award

仁寶賽道 · AI × 醫療照護

提 案 計 劃 書

PANCREASaver® 助胰見®

基於邊緣運算驅動之(SaMD)AI 胰臟癌偵測系統

參賽組別：組別一 — 臨床診斷與醫療決策創新

提案單位：仲智數位健康股份有限公司（PanCAD.ai Co., LTD）

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團隊代表：楊凱鈞　行銷業務總監

聯絡信箱：kc.yang@pancad.ai　　手機：0967-103-155

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摘　　要

胰臟癌為全球最致命惡性腫瘤之一，逾八成病患確診時已屬晚期，五年存活率長年低於一成。關鍵瓶頸在於小於二公分之早期病灶於常規電腦斷層（CT）中約四成為肉眼漏診。然而，**若能於腫瘤未達二公分時偵測並積極治療，五年存活率可自不足一成提升至約八成。** PANCREASaver® 助胰見® 為全球首創之全自動化(SaMD)AI 胰臟癌 CT 輔助偵測系統，整合深度學習與影像組學雙引擎，於 CT 掃描當下即時感知微小病灶，並主動驅動終端醫學影像資訊系統 PACS，回饋檢測結果、使 PACS 工作站同步載入高風險影像、自動產出結構化報告——實現「感知→決策→行動」之閉環。**系統對小於二公分腫瘤敏感度達 92.1%，全國多中心 1,473 例驗證 AUC 達 0.95，並成功揪回 12 名放射科醫師漏診癌中 11 名。**

本系統已取得台灣衛福部醫療器材許可證（衛部醫器製字第 007946 號）及美國 FDA Breakthrough Device Designation 突破性醫材認證，榮獲北美放射學會(Radiological Society of North America, RSNA)最高榮譽 RSNA Margulis Award（台灣首次獲獎）等九項國內外大獎，並佈局台美 10 件發明專利。系統已於臺大醫院、輔大醫院、博田國際醫院實際部署，累積 300+ 人次，通過聯新國際醫院 AI 信任中心 50 例第三方驗測。FDA 510(k) 已於 2026 年 7 月遞交申請。

本提案對準 2026 AI 創新獎「仁寶賽道——臨床診斷與醫療決策創新」組別，完全契合競賽核心宗旨：「**AI 是否直接影響設備行為或實體決策，而非僅分析資料**」。以下各章節依競賽評分標準（場景痛點 30% / 技術創新 25% / 硬體整合 25% / 商業模式 20%）逐項展開，以完整證據鏈證明 PANCREASaver® 已從「分析資料的被動軟體」進化為「主動警示胰臟癌病灶位置，驅動輔助醫療行為的(SaMD)AI 胰臟癌檢測系統」，尤其對於早期胰臟癌檢測之敏感度高達 92.1%，於臨床意義上早期發現，及早處置，讓癌王無所遁形，是仁寶賽道最具獲獎實力的提案。

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第一章　提案整體說明

〔評分構面：綜合呈現〕

★ 本節得分亮點　全章圍繞(SaMD)AI 核心主張：AI 不止分析資料，更直接驅動設備行為。以真實臨床案例破題，五項國際期刊證據鏈支撐，已取證、已落地、已救人。

一位７０餘歲的女性，因持續性黃疸就診後，電腦斷層僅顯示膽管擴張，未見明確腫瘤、內視鏡超音波導引下之穿刺切片判讀為陰性。若依循常規，她極可能被歸為良性而僅作後續觀察——然而，這正是胰臟癌奪走無數生命的典型情況。經 PANCREASaver® 重新分析，系統判定為陽性並標示胰頭可疑病灶，最終手術確診，及時挽回一命。

圖 1 臨床案例－ 原始影像

圖 2 臨床案例－ PANCREASaver® AI 檢測後影像

胰臟癌之所以被稱為「癌王」，在於其早期幾無症狀、病程進展迅速，逾八成病患確診時已屬晚期，五年存活率低於百分之十。電腦斷層掃描雖為主要檢測工具，然對小於二公分之早期病灶，臨床漏診率高達四成——此非個別醫療疏失，而是人眼辨識之客觀極限 [R1][R4]。然而多項研究已證實，若能於腫瘤仍小於二公分時偵測並施以積極治療，五年存活率可顯著提升至約八成 [R2]。

PANCREASaver® 助胰見® ——一套由臺灣自主研發、全球首創之(SaMD) "助胰見" 人工智慧胰腺癌輔助偵測系統 ("PANCREASaver" Image Analytics Software)，正是針對此一醫療現場痛點而生。其結合深度學習（CNN）與影像組學（Radiomics）雙引擎，直接讀取標準 DICOM 影像，於掃描完成之際即時感知病灶，並主動驅動終端硬體——觸發閱片工作站實體警示、自動產出標記病灶影像、自動載入高風險序列於 PACS。系統對小於二公分早期腫瘤之敏感度達 92.1%，並研究證實成功偵測放射科醫師漏診病例中之 11/12 例（92%）；全國性真實世界研究（1,473 例）敏感度 89.7%、特異度 92.8%、AUC 0.95 [R1]。正因臨床價值明確，本產品獲美國 FDA 突破性醫材（美國 FDA「突破性醫療器材計畫」(Breakthrough Devices Program)。專為能更有效治療或診斷危及生命之嚴重疾病的創新技術而設立。獲此認定後，廠商可享有與 FDA 優先溝通、提前審查等資源，以大幅縮短上市時間。）認定，並已取得衛福部 TFDA 醫材許可（衛部醫器製字第 007946 號），為臺灣首張胰臟癌 AI 醫材許可證。

核心效能指標	數值	驗證情境
<2cm 早期腫瘤敏感度	92.1%	國內資料集
AI 揪回醫師漏診病例	11 / 12 (92%)	國內資料集
AI vs 放射科醫師 敏感度	98.3% vs 92.9%	兩國內資料集合併
全國真實世界 敏感度	89.7%	全國 1,473 例
全國真實世界 特異度	92.8%	全國 1,473 例
全國真實世界 AUC	0.95	全國 1,473 例
傳統 CT 對 <2cm 腫瘤漏診率	約 40%	臨床現況（對照）

表 1　核心效能指標與頂尖學術期刊對照

本系統之定位並非取代醫師，而是以 AI 於臨床判讀前先行篩選，將最可疑之影像優先呈現給醫師，最終診斷仍由醫師裁定——賦予醫師一雙不受疲勞影響的輔助之眼。就流行病學而言，胰臟癌雖非發生率最高之癌別，然其高致死特性與臺灣每年約 90 萬人次之腹部 CT 檢查

量，意謂大量早期病灶潛藏於既有影像中。每提升一分早期檢出率，即多挽回一名病患及一個家庭。此即本系統存在之根本價值——讓「早期發現」不再仰賴機運，而成為每一次掃描之常態。

第二章　團隊組成與分工

〔評分構面：技術創新 25%〕

★ 本節得分亮點　產學研鐵三角：臺大演算法 × 臺大醫院臨床 × 仲智數位健康 PanCAD.ai 商轉。九項大獎、五篇頂刊、十件專利、五組核心團隊跨越臨床 / 數學 / 資工 / 法規 / 商務，五年不間斷推進——這是後進者難以複製的系統性壁壘。

如同過去的醫療革命從非單一領域所能獨立成就，PANCREASaver® 之所以能自實驗室走入臨床，關鍵在於學研、臨床與商轉三方之深度整合。

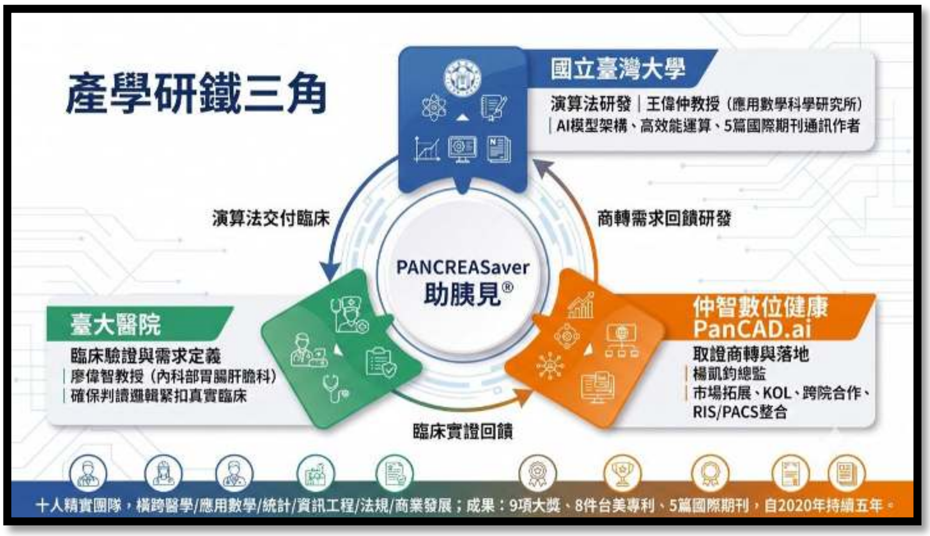


圖 3 產學研團隊由國立臺灣大學，國立臺灣大學附設醫院，仲智數位健康股份有限公司組成

技術核心：王偉仲教授（國立臺灣大學應用數學科學研究所）及其團隊

主導 AI 模型架構設計與高效能科學計算，為團隊五篇國際期刊之通訊作者。專精於深度學習、數值模擬與高維度資料分析，奠定 PANCREASaver® 演算法之全球領先地位。其領導之 MeDA Lab 為台灣頂尖之醫學影像 AI 實驗室。

臨床驗證與需求定義：廖偉智教授（臺大醫院內科部胃腸肝膽科主治醫師暨綜合診斷部主任）及其團隊

以第一線消化醫學專業，確保 AI 之判讀邏輯緊扣真實臨床情境，使技術不致淪為缺乏實用性之工具。負責臨床研究設計、真實世界驗證與多專科協作，為所有五篇臨床論文之共同通訊作者。

商業化與落地推進：仲智數位健康股份有限公司（PanCAD.ai）

楊凱鈞總監統籌市場拓展、KOL 導入與跨院合作。PanCAD.ai 系統工程團隊專責邊緣設備串接、RIS/PACS 底層整合與資安合規治理。團隊橫跨臨床醫學、應用數學、資訊工程、法規與商業發展等領域，形成「臺大演算法研發 → 臺大醫院臨床驗證 → PanCAD.ai 取證商轉」之產學研鐵三角。

此一協作模式之成效已具體反映於其成果：累計榮獲國內外九項指標大獎、佈局台美十件發明專利，並自 2020 年起持續發表五篇同儕審查研究於國際頂尖期刊。團隊歷經全國性資料驗證、國際法規認證與臨床商轉，五年間未曾間斷推進產品落地。我們深信，此一難以複製之整合能力，正是 PANCREASaver® 能持續領先、並將學術成果實質轉化為病患福祉之根本所在。

表 2　團隊核心成員與分工

姓名	角色	所屬單位	職稱	核心貢獻
王偉仲	技術 / 共同創辦人	國立臺灣大學	應數所教授	AI 模型架構、科學計算、5 篇通訊作者
廖偉智	臨床 / 共同創辦人	國立臺灣大學附設醫院	綜合診療部主任 / 教授	臨床驗證、需求定義、多專科協作
楊凱鈞	團隊代表 / 總監	仲智數位健康股份有限公司 PanCAD.ai Co., Ltd.	總監	市場拓展、KOL 導入、跨院合作商轉
PanCAD.ai 工程團隊	系統研發與部署	仲智數位健康股份有限公司 PanCAD.ai Co., Ltd.	10 人精實團隊	RA 取證，邊緣運算、RIS/PACS 整合、資安合規

第三章　產業場景痛點

〔評分構面：場景痛點 30%——最高權重〕

★ 本節得分亮點　三大痛點直擊評分核心：（1）CT 設備純被動擷取，無主動預警——呼應「設備行為」評分；（2）放射科人力結構性過載——呼應「實務需求」；（3）早期漏診＝錯失 80% 存活率——數字直接證明問題嚴重性。每一痛點皆附文獻證據。

試想：病患得以存活的關鍵資訊，其實早已存在於其 CT 影像之中；然而，僅因早期胰臟癌病灶小於二公分而未能為肉眼所辨識，造成錯失及早治療的契機。此即胰臟癌臨床診斷中最令人扼腕之處，亦為本提案所欲根本解決之問題：

痛點一：設備之被動性——CT 只能拍，不能警示

現行 CT 設備僅具影像擷取功能，缺乏主動預警機制，無法於掃描當下提示可疑病灶；放射科醫師須於數百張切片中憑經驗逐一判讀。文獻顯示，傳統 CT 對小於二公分早期腫瘤之漏診率高達約四成，51.7% 的 CT 影像在事後回顧發現胰臟病變跡象 [R1][R4]。此非個別的醫療疏失，而是人眼辨識之客觀極限。關鍵在於：現有設備完全被動，無法在擷取影像的當下驅動任何預警行為——這正是 PANCREASaver® AI 最能發揮價值的切入點。

痛點二：醫療負荷之結構性壓力——系統性困境，非經驗可解

放射科醫師工作量龐大，臺灣每年約 90 萬人次腹部 CT 檢查，平均每次檢查約５００張 DICOM 切片影像．早期不明顯之微小病灶於高強度判讀環境下極易被忽略。此為系統性而非可仰賴經驗能克服的困境——再資深的醫師也無法保證每一份影像的每一張切片都被同等仔細地審視。

痛點三：臨床診斷之多重侷限——檢查做到極限仍可能偽陰性

胰臟癌病患往往須反覆接受內視鏡超音波、穿刺切片等侵入性檢查，不僅造成身心負擔，且仍可能出現偽陰性結果，使診斷陷入僵局。本團隊已累積多例活檢陰性但 AI 正確標示病灶的臨床案例——當傳統診斷路徑走到盡頭時，AI 提供的那一個額外判斷，可能就是救命的關鍵。上述問題之代價極為沉重。臨床證據明確指出：於腫瘤小於二公分時及早偵測並積極治療，五年存活率可自個位數大幅提升至八成以上 [R2]。

此一痛點並非臺灣所獨有——胰臟癌於美國為第三大癌症死因且持續上升，全球放射醫學界普遍面臨人力短缺與早期漏診之共同挑戰。因此，臨床現場真正需要的是一套能於掃描當下

主動偵測、即時驅動警示，且不增加醫師負擔之(SaMD)AI 預警系統。本團隊提案所欲建立的，正是從根本改變胰臟癌篩檢典範——使被動影像為主動預警系統，讓早期發現成為制度化常態。

第四章　PANCREASaver® AI 解決方案

〔評分構面：技術創新 25% ＋ 硬體整合 25% — 合計 50%〕

★ 本節得分亮點　這是全案最關鍵的章節——直接回應競賽靈魂考題：「AI 是否直接影響設備行為或實體決策」。PANCREASaver 完整實現「感知→決策→驅動」閉環，每一步都有文獻支持、有臨床案例佐證、有已部署硬體為證。

PANCREASaver® 之根本差異在於：其 AI 判讀結果可直接驅動醫院 PACS, RIS 系統，主動介入臨床流程——而非僅止於產出一份靜態報告等待醫師有空時翻閱。此即其符合本產品定義之核心：AI 不只分析資料，更驅動 PACS 系統 [R13]。

感知—決策—驅動 三層閉環架構

【感知層】即時感測，零人工介入

系統部署於醫院端邊緣 GPU 伺服器；病患完成 CT 掃描後，DICOM 影像透過標準 C-STORE 協定即時串流進入 AI 推論管線，全自動完成影像分析與推論、器官 ROI 定位與去識別化前處理。全程無須醫療人員手動圈選器官——這是臨床採用的最低門檻要求。

【決策層】雙引擎協同，精準鎖定微小病灶

二維與三維卷積神經網路（改良 3D ResNet + Attention Gate）結合影像組學（Radiomics）逾百維度特徵（一階統計、形狀、GLCM、GLRLM、小波特徵），經集成模型融合判讀，於數分鐘內鎖定小於二公分之可疑病灶並輸出腫瘤位置與風險機率。雙引擎設計，專為提升對最易漏診之微小病灶的偵測敏感度——經 Lancet Digital Health 及 Radiology 等頂尖期刊驗證，對 <2cm 腫瘤敏感度達 92.1% [R1]。

【驅動層】AI 決策轉化為設備行為（PANCREASaver® AI 的關鍵行動層）

一旦判定為高風險，系統除產出報告外，更透過標準醫療通訊協定（HL7 / FHIR / DICOM SC）與醫院 RIS 及 PACS 進行硬體與資料層級之指令連動——觸發閱片工作站發出實體警示、自動產出標記病灶影像、優先載入高風險序列於 PACS 閱片清單頂端，主動引導醫師視線。這正是 PANCREASaver® AI 與一般診斷軟體的本質區別：AI 的決策直接驅動 PACS 系統。

臨床案例實證

案例一：一名七十餘歲女性，新發糖尿病併 CA 19-9 輕度上升，CT 肉眼未見明確腫瘤，經 PANCREASaver® 判定陽性並標示胰頸可疑區域，術後病理確診為 1.8 公分早期胰臟腺癌（PDAC）。此類個案於傳統流程中極易被漏診而錯失手術良機。

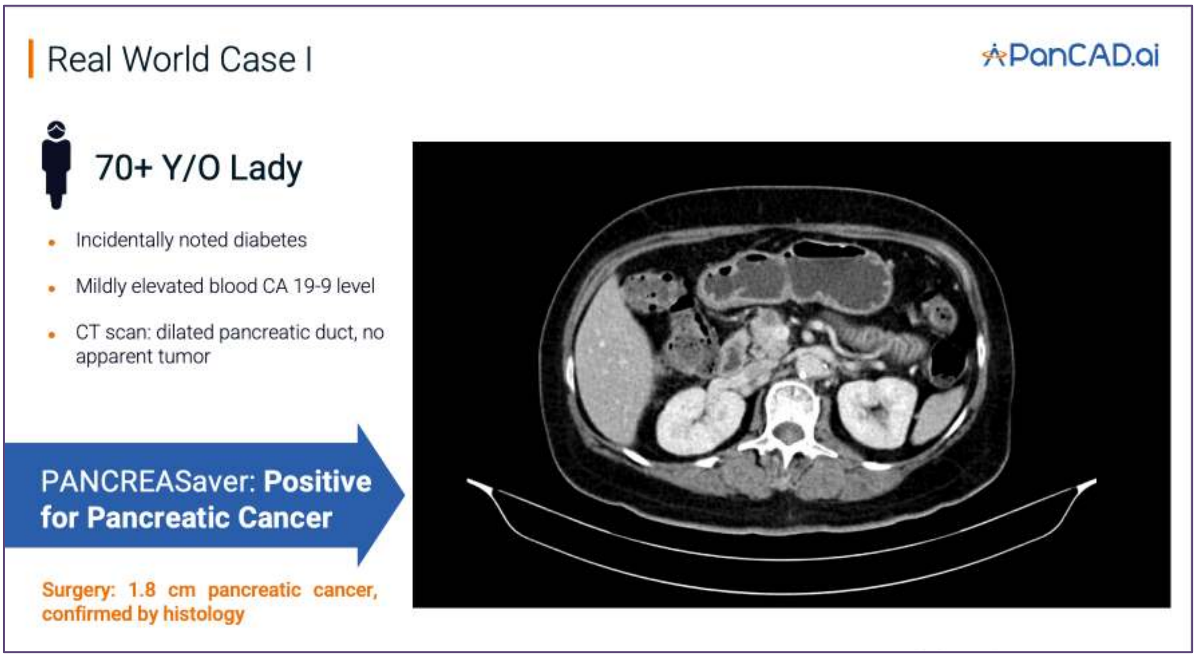


圖 4 案例一　突破肉眼盲區：攔截無明顯腫塊之早期胰臟癌-1

圖 5 案例一　突破肉眼盲區：攔截無明顯腫塊之早期胰臟癌-2

案例二：一名六十餘歲女性，出現進行性黃疸、CT 顯示膽管擴張而無明顯胰腺腫瘤，內視鏡超音波及活檢均無診斷性結論，但透過 PANCREASaver® 診斷為陽性並顯示胰頭可疑區域，最終手術切除後經病理證實為 PDAC。此案例特別凸顯了 AI 在傳統診斷路徑走到盡頭時的獨特價值。

綜言之，本系統將「被動之診斷軟體」轉化為「由 AI 主動驅動臨床醫療」，於不改變醫師既有閱片習慣之前提下，實現決策驅動實體行為之變革——使每一次 CT 掃描皆成為主動攔截早期病灶的契機。

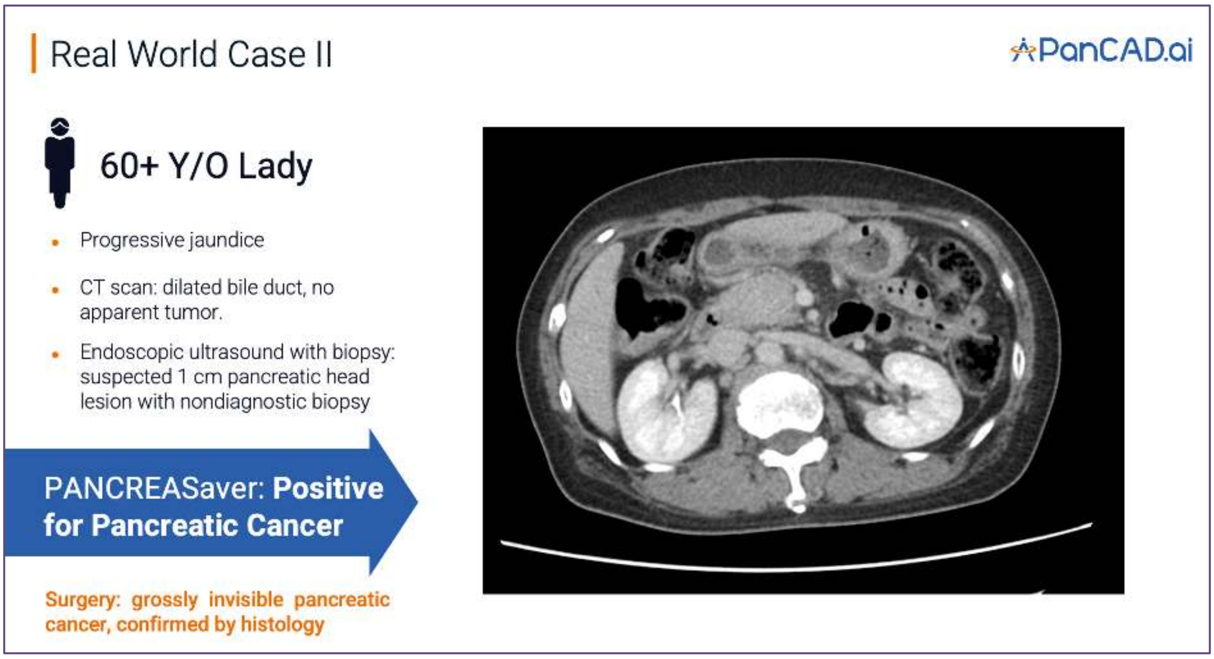


圖 6 案例二　克服侵入性檢查偽陰性：精準定位 1 公分極早期腫瘤-1

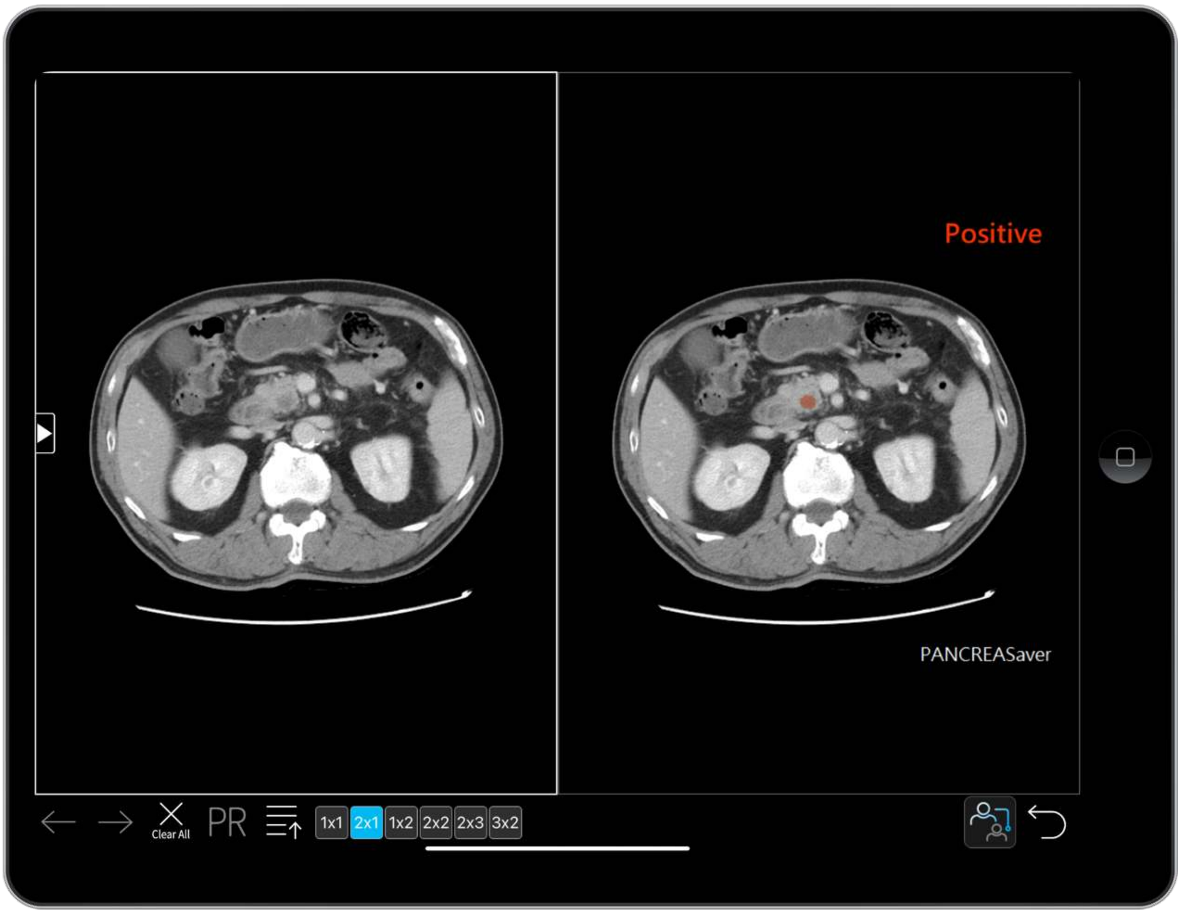
圖 7 案例二　克服侵入性檢查偽陰性：精準定位 1 公分極早期腫瘤-2

第五章　具體任務與複審 Demo 規劃

〔評分構面：評分構面：硬體整合 25%（複審現場 DEMO）〕

★ 本節得分亮點　六項可 Demo 任務全部對準複審評分——含「環境干擾與隨機變數測試」準備。現場實機運行，不是放影片。已備妥 Siemens / GE 雙機型案例及偽陽性挑戰陰性案例。

為驗證系統之實際效能，本團隊將於複審階段進行現場實機示範，展示 AI 判讀。具體可示範之六大任務如下：



任務一：即時驅動邊緣運算核心模組

模擬 CT 斷層掃描結束後進行傳輸影像，邊緣伺服器接收 CT 影像後，成功觸發邊緣運算伺服器主機，於數分鐘內辨識 <2cm 病灶，成功觸發回傳 AI 辨識結果，驗證 AI 輔助決策功能。

任務二：自動報告產出與系統驅動

同步傳輸至 PACS，將辨識結果模擬在醫院端 PACS 系統自動產出 AI 結果，並自動產出 RIS 報告摘要，過程為自動化無因導入 AI 系統而變更現有醫院既有工作流。

任務三：病灶可視化標示

系統展是，系統並列呈現原始影像與 AI 標註影像，以高對比視覺標示腫瘤位置，除引導醫師判讀外，亦有助於醫病溝通。

▶ 任務四：臨床流程無縫整合

於完全不改變醫師既有閱片習慣之前提下，驗證系統即插即用之特性——此為醫師長期採用之關鍵。

▶ 任務五：真實案例重現

以去識別化之真實個案示範：1.8 公分早期 PDAC 及活檢偽陰性 1 公分病灶，展示 AI 如何攔截傳統檢查難以察覺之早期病變。

▶ 任務六：效能實測與隨機變數測試

現場展示單份逾五百張切片之 CT 於五分鐘內完成推論。另備多個不同 CT 影像案例含陽性與陰性案例。

上述任務皆非實驗室假設，而是系統於臺大醫院、輔大醫院實際運作之常態：截至 2026 年 7 月，累計臨床輔助判讀達 300+人次。我們能以真實個案重現完整臨床流程——影像接收、判讀、警示觸發與報告產出，每一環節皆為導入院所的例行運作。

我們深信：令評審親眼見證 AI 偵測病灶、工作站觸發警示、報告自動產出之完整歷程，遠較任何簡報論述更具說服力。這一刻的親眼見證，勝過千言萬語。

第六章　系統架構與 AI 整合設計

〔評分構面：技術創新 25% ＋ 硬體整合 25%〕

★ 本節得分亮點　三層架構（邊緣感知→AI 決策→醫療系統驅動）完整呈現 (SaMD)AI 的工程實現。PANCREASaver Bridge 中介軟體為自研核心。S-SDLC 安全開發、去識別化、LDAP/AD 整合——資安合規一次到位。

醫療現場對系統之要求，在於判讀精準、反應即時，且能安全地於院內邊緣運算環境穩定運行。為此，本系統採全程隱身於醫院封閉內網之設計，不依賴外部網路，確保最高等級之資訊安全。

三層式邊緣運算架構

一、邊緣感知層（Perception）

CT 產出 DICOM 影像後，透過標準 DICOM C-STORE 協定直連院內邊緣運算 GPU 伺服器（NVIDIA RTX 4090 / A1000），全自動完成演算法運算、器官 ROI 定位與去識別化。感測對象為標準化 CT 影像序列，無須醫療人員手動介入。亦支援雲端部署方案供未來擴充。

二、AI 決策層（Decision）

二維與三維卷積神經網路（改良 3D ResNet + Attention Gate）結合影像組學逾百維度特徵，經集成模型融合判讀。模型於全國性真實世界資料完成驗證：敏感度 89.7%、特異度 92.8%、AUC 0.95 [R1]。以真實世界資料——而非實驗室精選資料——進行驗證，對臨床信任至關重要。

三、醫療系統驅動層（Action）

判定陽性後，透過標準醫療通訊協定（HL7 / FHIR / DICOM SC）與內網交換指令，同步完成：報告產出 → RIS 寫入 → PACS 畫面整合與警示。PANCREASaver Bridge 為自研核心中介軟體，負責各層間的訊息路由、事件日誌與異常重試機制。提供醫師審閱與簽署介面。

資訊安全與合規治理

系統導入安全軟體開發生命週期（S-SDLC）；病患影像於推論前完成端點去識別化；遵循《個人資料保護法》及衛福部負責任 AI 治理框架；使用者身分驗證透過 LDAP / AD 整合院內帳號系統並具完整操作稽核日誌。

技術壁壘

以台美１０件發明專利佈局（台灣 6 件 + 美國 4 件），採系統級與方法級雙重權利範圍撰寫，確認於臺灣、美國、歐盟及中國均無侵權風險。整體架構之核心優勢在於：完全融入放射科醫師既有工作流程——易於介接、推論迅速、資料安全。此三項條件之達成，方為醫療 AI 得以進入臨床並長期留存之前提。

號	專利名稱（中 / 英）	台灣專利號	美國專利號	取證時間（台 / 美）
1	醫療影像分析系統及其方法 *Medical Image Analyzing System and Method Thereof*	**TWI745940**	**US 11,424,021**	2021.11 / 2022.08
2	醫療影像分析系統及其方法 *Medical Image Analyzing System and Method Thereof*	**TWI767439**	**US 11,684,333**	2022.06 / 2023.06
3	醫療影像分析系統及其方法 *Medical Image Analyzing System and Method Thereof*	**TWI776716**	**US 12,067,717**	2022.09 / 2024.08
4	醫療影像分析系統及其方法（以機器學習識別胰臟癌） *Medical Image Analyzing System for Identifying Pancreatic Cancer Using Machine Learning and Method Thereof*	**TWI790788**	**US 12,106,473**	2023.01 / 2024.10
5	進階影像分析系統、方法及其電腦可讀媒介 ADVANCED IMAGE ANALYSIS SYSTEM, METHOD AND COMPUTER READABLE MEDIUM THEREOF	**TWI918507**		2026.03
6	進階影像分析系統、方法及其電腦可讀媒介 ADVANCED IMAGE ANALYSIS SYSTEM, METHOD AND COMPUTER READABLE MEDIUM THEREOF	**TW202609795A**		2026.03

第七章　硬體平台與實體驗證規劃

〔評分構面：硬體整合 25%——關鍵得分章節〕

> **★ 本節得分亮點　已部署、已上線、已驗證——不是「規劃中」或「預計」。臺大＋輔大＋博田三家醫院實際運行，190 人次，聯新 AI 信任中心 50 例第三方驗測，FDA 510(k) 已遞交。用真實世界的運行數據回應「實體驗證」評分。**

任何無法於真實臨床環境落地之技術，終究僅為實驗室之理論。本團隊之驗證策略，係將邊緣運算硬體直接部署於臨床第一線，於真實病患與實際流程中接受檢驗。

硬體平台規格

- 高效能 GPU 邊緣運算主機（NVIDIA RTX 4090 / A1000 等級以上），單份逾五百張切片 CT 影像約可於五分鐘內完成推論
- 內建實體警示模組（警示燈號 / 工作站 API 聯動）
- 標準醫療網路介接協議（DICOM / HL7 / FHIR），與院內既有 PACS 及 RIS 無縫整合
- 作業系統：Linux or Windows with GPU
- PANCREASaver® Bridge 中介軟體（自研）：訊息路由、事件日誌、異常重試機制

實體驗證四階段

第一階段｜影子測試（已完成）：

團隊已於頂尖醫學中心之後台同步運行系統，在完全不干擾既有醫療流程之前提下，驗證 AI 驅動硬體警示之速度、準確度與穩定度。

第二階段｜真實世界營運驗證（已完成並持續進行）：

系統已於臺大醫院總院、輔大醫院聖路加健檢中心正式上線，截至 2026 年 7 月累計臨床輔助判讀 300+人次；博田國際醫院 2026 年 6 月起導入；聯新國際醫院透過衛福部「負責任 AI 執行中心」進行 50 例第三方驗測，預計於 2026 年 6 月導入；彰化基督教醫院採購程序完成，預計於 2026 年 8 月導入。此等由院方主動發起之導入與驗測，印證硬體落地之穩定性與臨床價值。

第三階段｜持續監控與前瞻性研究（進行中）：

臺大研發團隊持續監測模型於真實硬體環境下之靈敏度；規劃 2026 Q3–2027 Q2 於 5 家以上區域醫院進行 500+ 例前瞻性臨床研究。

第四階段｜法規驗證（已完成 TFDA＋進行中 FDA）：

系統已通過 TFDA 醫材查驗登記（衛部醫器製字第 007946 號），為臺灣首張胰臟癌檢測 AI 醫材許可證；美國 FDA 510(k) 已於 2026 年 7 月遞交申請，以國際最高標準再次驗證軟硬體整合之安全性與有效性。

將技術直接推向真實臨床戰場，是最誠實亦最嚴苛之驗證方式。本團隊選擇此途，正因唯有如此，方能對得起每一位將生命託付於醫療現場之病患。已上線、已實際輔助臨床、已推進國際認證——此即本團隊對「落地」二字最務實之交代。

表 3　PANCREASaver® 實際部署進度

部署機構	狀態	累積案例	備註
臺大醫院總院	✅ 正式上線	35 人次	放射科即時 AI 輔助判讀
輔大醫院聖路加健檢中心	✅ 正式上線	180 人次 （年度預約 400 例）	臺灣最大腹部 MRI 健檢中心
博田國際醫院	✅ 2026.06 導入	11 例（預約）	南部市場擴張首站
聯新國際醫院	🔄 驗測中 📋 部署中	50 例	衛福部 AI 信任中心第三方驗測
彰化基督教醫院	📋 部署中	—	中部市場拓展目標
部立台北醫院	採購程序中	—	部立醫院指標
中心綜合醫院	採購程序中	—	北市高階健檢

（高階醫學影像中心）			
其他北中南醫院 （6間機構）	採購程序中	—	—

第八章　商業模式與市場發展規劃

〔評分構面：商業模式 20%〕

★ 本節得分亮點　SaaS 年約訂閱制。臺灣 90 萬人次 / 年腹 CT × 全球 130 億美元市場。已取得 Term Sheet、建置 QMS、聯新加速器資源。後進者壁壘：率先研發→率先臨床→率先取證。

本產品之市場拓展已具實質進展，而非停留於規劃階段。

導入實績

系統已導入臺大醫院、彰化基督教醫院、輔大醫院聖路加健檢中心；博田國際醫院為南部擴張首站（2026 年 6 月起）；聯新國際醫院透過衛福部「負責任 AI 執行中心」進行 50 例第三方驗測；彰化基督教醫院議價中。截至 2026 年 7 月累計臨床輔助判讀 300+人次。但胰臟癌不是常見癌別，每一次判讀都可能攔截一條命，我們看的不是絕對量，是成長斜率：逐月攀升。彰化基督教醫院、聯新國際醫院目前正部署中，另有多家機構正進行採購程序中．

商業模式：雙軌策略

【SaaS 年約訂閱制】

依 CT 掃描量分級計費。醫院導入後可新增針對高致死率癌症之自費篩檢項目，於不增加放射科醫師工作量之前提下創造新收益。早期發現可避免晚期龐大之化療、標靶與住院支出，實質為病患家庭與國家健保節省成本。

【SaaS 按次訂閱制】

為加速市場滲透並減輕醫療機構的預算負擔，我們採取將 PANCREASaver® 預先載入至 Edge AI 伺服器 / 醫規工作站的策略。這種「硬體 + AI 服務」的綑綁銷售（Bundle Sales）結合按次計費機制，讓醫院能以極低的風險無痛導入頂尖 AI 應用。此模式不僅能快速提高市場接受度，更能與硬體供應鏈產生強大的戰略綜效。

市場規模

臺灣每年約 90 萬人次接受腹部 CT 檢查；全球 AI 醫學影像市場預估於 2030 年達 130 億美元，年複合成長率逾 35%。PANCREASaver 之 FDA 510(k) 已遞交，規劃 2027 年啟動美國初步商業化、2028 年歐盟 CE MDR 申請。後續研發將推進無顯影劑 CT（NC-CT）與磁振造影（MRI）之新模組，擴大適應症覆蓋。

公司永續與競爭壁壘

圖 8 國內外指標大獎：台灣之光的權威背書

本產品核心技術於 2023 年榮獲北美放射學會（RSNA）Alexander R. Margulis Award——放射醫學界最高科學榮譽，台灣團隊首度獲此殊榮，並取得美國 FDA 突破性醫材認定與台灣 TFDA 許可證。國內更囊括 2025 國家藥物科技研究發展獎金質獎、2024 SNQ 國家生技醫療品質獎銀獎、2024 台北生技獎金獎，以及國家新創獎多屆肯定

現有十人精實團隊、已取得投資意向書（Term Sheet），並正建置自主醫療器材品質管理系統（QMS），預計 2026 Q4–2027 H1 取證，屆時將具備完整醫材製造能力；另獲 2026 聯新數位健康加速器資源挹注。

於胰臟癌 AI 此一領域，率先完成研發、率先進入臨床、率先取得國際認證者，將建立後進者難以跨越之門檻。本團隊已領先數年——兩家醫學中心導入、300+ 人次實證、九項國內外大獎、10 件發明專利及五篇國際頂尖期刊——均非後進者短期所能補足。這不只是一具商業爆發力的產品，更是一項足以挽救眾多生命的志業。

評分構面對照總表

以下對照表呈現本提案各章節與競賽四大評分構面之對應關係，供評審快速查閱：

表 4　評分構面與對應章節一覽

評分構面	權重	對應章節	PANCREASaver 得分關鍵
場景痛點與解決方案	30%	第一章、第三章	三層痛點（設備被動 / 人力過載 / 診斷侷限）→(SaMD) AI 主動偵測篩檢；<2cm 敏感度 92.1%；AUC 0.95
技術創新與系統架構	25%	第四章、第六章	感知→決策→驅動閉環；CNN + Radiomics 雙引擎；5 篇頂刊驗證；台美 8 件專利佈局
硬體整合與實作計畫	25%	第五章、第七章	已部署 5 家醫院實際運行 300+人次；6 項可 Demo 任務含隨機變數測試；
商業模式與市場性	20%	第八章	SaaS 訂閱制服務，台灣已上市，美國 FDA 已遞件申請

參考文獻

[R1] ★ Chen PT, Chang D, Wu T, et al. "Pancreatic Cancer Detection on CT Scans with Deep Learning: A Nationwide, Real-World Study." Radiology, 2023; 308(2): e220152. DOI: 10.1148/radiol.220152. PMID: 36926666.

※ PANCREASaver 核心臨床驗證文獻。

[R2] ★ Egawa S, Takeda K, et al. "Clinicopathological aspects of small pancreatic cancer." Pancreas, 2004; 28(3): 235-240. DOI: 10.1097/00006676-200404000-00004.

※ <2cm 胰臟癌積極治療，五年存活率可達 58–80%。

[R3] Sung H, Ferlay J, et al. "Global Cancer Statistics 2020: GLOBOCAN." CA: A Cancer Journal for Clinicians, 2021; 71(3): 209-249. DOI: 10.3322/caac.21660.

※ 全球癌症統計。

[R4] ★ Hoogenboom SA, et al. "Prevalence, features, and explanations of missed and misinterpreted pancreatic cancer on imaging." Abdominal Radiology, 2022; 47(12): 4160–4172. DOI: 10.1007/s00261-022-03671-6.

※ 51.7% CT 事後發現胰臟病變跡象被遺漏。

[R5] ★ Mukherjee S, et al. "Radiomics-based ML Models Can Detect Pancreatic Cancer on Prediagnostic CT at a Substantial Lead Time." Gastroenterology, 2022; 163(5): 1234-1245. DOI: 10.1053/j.gastro.2022.07.015.

※ AI 可在臨床診斷前 386 天偵測亞臨床胰臟癌。

[R6] Cao K, et al. "Large-scale pancreatic cancer detection via non-contrast CT and deep learning." Nature Medicine, 2023; 29: 3033–3043. DOI: 10.1038/s41591-023-02640-w.

※ 非對照 CT 大規模偵測胰臟癌可行性。

[R7] Liu KL, Wu TH, Chen PT, et al. "Deep learning to distinguish pancreatic cancer from benign pancreatic conditions on CT." Radiology: Imaging Cancer, 2020; 2(4): e190096. DOI: 10.1148/rycan.2020190096.

※ PANCREASaver 前期方法論。

[R8] Liao WC, Chen WY, et al. "AI detects missed pancreatic cancer at abdominal CT." The Lancet Digital Health, 2021; 3(6): e346-e347. DOI: 10.1016/S2589-7500(21)00060-X.

※ AI 回溯性揪出漏診胰臟癌。

[R9] Chen WY, Liao WC, et al. "Radiomics-based ML for differentiating pancreatic cancer from mass-forming chronic pancreatitis on CT." BMC Cancer, 2023; 23: 345. DOI: 10.1186/s12885-023-10777-5.

※ 放射組學於胰臟癌鑑別診斷。

[R10] Yao L, et al. "A review of deep learning and radiomics approaches for pancreatic cancer diagnosis." Current Opinion in Gastroenterology, 2023; 39(5): 436–447. DOI: 10.1097/MOG.0000000000000966.

※ 深度學習與放射組學於胰臟癌診斷綜述。

[R11] Lopez-Ramirez F, et al. "Early detection of pancreatic cancer on CT: advancements with deep learning." Radiology Advances, 2025; 2(5): umaf028. DOI: 10.1093/radadv/umaf028.

※ 深度學習於胰臟癌早期偵測最新進展。

[R12] Pongprasobchai S, et al. "Long-term survival and prognostic indicators in small pancreatic cancer." Pancreatology, 2008; 8(6): 587-592. DOI: 10.1159/000161009.

※ <2cm 胰臟癌長期存活與預後指標。

[R13] Ferrone CR, et al. "Pancreatic adenocarcinoma: the actual 5-year survivors." J Gastrointest Surg, 2008; 12(4): 701-706. DOI: 10.1007/s11605-007-0384-8.

※ 胰臟癌實際五年存活者分析。

附件

本大獎採實證導向，參獎單位若於申請書提及之各項專業實績，均須於本區檢附對應之證明文件影本。

1、 相關證明文件

1. 專利文件

序號	專利名稱（中／英）	台灣專利號	美國專利號	取證時間（台／美）
1	醫療影像分析系統及其方法 *Medical Image Analyzing System and Method Thereof*	**TWI745940**	**US 11,424,021**	2021.11 / 2022.08
2	醫療影像分析系統及其方法 *Medical Image Analyzing System and Method Thereof*	**TWI767439**	**US 11,684,333**	2022.06 / 2023.06
3	醫療影像分析系統及其方法 *Medical Image Analyzing System and Method Thereof*	**TWI776716**	**US 12,067,717**	2022.09 / 2024.08
4	醫療影像分析系統及其方法（以機器學習識別胰臟癌） *Medical Image Analyzing System for Identifying Pancreatic Cancer Using Machine Learning and Method Thereof*	**TWI790788**	**US 12,106,473**	2023.01 / 2024.10
5	進階影像分析系統、方法及其電腦可讀媒介 ADVANCED IMAGE ANALYSIS SYSTEM, METHOD AND COMPUTER READABLE MEDIUM THEREOF	**TWI918507**		2026.03
6	進階影像分析系統、方法及其電腦可讀媒介 ADVANCED IMAGE ANALYSIS SYSTEM, METHOD AND COMPUTER READABLE MEDIUM THEREOF	**TW202609795A**		2026.03

- 醫療影像分析系統及其方法/MEDICAL IMAGE ANALYZING SYSTEM FOR IDENTIFYING PANCREATIC CANCER USING MACHINE LEARNING AND METHOD THEREOF

I790788

(19)中華民國智慧財產局

(12)發明說明書公告本　(11)證書號數：TW I790788 B

(45)公告日：中華民國 112 (2023) 年 01 月 21 日

(21)申請案號：110139305　(22)申請日：中華民國 110 (2021) 年 10 月 22 日

(51)Int. Cl.：*G16H30/40 (2018.01)*　*G16H50/50 (2018.01)*
G06T7/00 (2017.01)　*G06N3/08 (2006.01)*

(30)優先權：2020/10/23　美國　63/104,568

(71)申請人：國立臺灣大學 (中華民國) NATIONAL TAIWAN UNIVERSITY (TW)
臺北市大安區羅斯福路 4 段 1 號

(72)發明人：王偉仲 WANG, WEI CHUNG (TW)；廖偉智 LIAO, WEI CHIH (TW)；劉高郎 LIU, KAO LANG (TW)；陳柏廷 CHEN, PO TING (TW)；王柏川 WANG, PO CHUAN (TW)；張大衛 CHANG, DA WEI (TW)

(74)代理人：張家彬

(56)參考文獻：

TW 202013309A　CN 107403201A
CN 111448584A　US 2020/0272864A1

審查人員：彭智煇

申請專利範圍項數：30 項　圖式數：2　共 29 頁

(54)名稱

醫療影像分析系統及方法

(57)摘要

本發明提供一種醫療影像分析系統及方法，主要將至少一病患影像輸入神經網路模組之第一模型，以得到具有確定該病患影像之臟器及腫瘤位置與範圍之結果，再將該結果分別輸入第一分析模組之第二模型及第二分析模組之第三模型，以得到對應該病患影像之至少一第一預測值及至少一第二預測值，最後根據該第一預測值及該第二預測值，輸出一判斷結果。本發明可自動化第一模型、第二模型及第三模型之間的流程，並有效增進胰臟癌之辨識率。

- 醫療影像分析系統及其方法/MEDICAL IMAGE ANALYZING SYSTEM AND METHOD THEREOF

I745940

(19)中華民國智慧財產局

(12)發明說明書公告本　**(11)證書號數：TW I745940 B**

(45)公告日：中華民國 110 (2021) 年 11 月 11 日

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(21)申請案號：109113320　　(22)申請日：中華民國 109 (2020) 年 04 月 21 日

(51)Int. Cl.：　***G16H30/00　(2018.01)***　　***G16H50/20　(2018.01)***

　　　　　　G06T1/40　(2006.01)

(30)優先權：2019/05/10　美國　62/845,922

(71)申請人：國立臺灣大學 (中華民國) NATIONAL TAIWAN UNIVERSITY　(TW)

臺北市大安區羅斯福路 4 段 1 號

(72)發明人：廖偉智 LIAO, WEI CHIH (TW)；王偉仲 WANG, WEI CHUNG (TW)；劉高郎 LIU, KAO LANG (TW)；陳柏廷 CHEN, PO TING (TW)；吳亭慧 WU, TING HUI (TW)

(74)代理人：陳孚竹；張家彬

(56)參考文獻：

TW	I587844	TW	201839128A
CN	107767362A	CN	109584252A
US	9486142B2		

審查人員：馮耀嘉

申請專利範圍項數：22 項　　圖式數：5　　共 36 頁

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(54)名稱

醫療影像分析系統及其方法

(57)摘要

本發明提供一種醫療影像分析系統及其方法，主要針對具有標記對應臟器位置之分割標籤之處理影像進行擷取，以產生複數個影像區塊，並可以複數個影像區塊對深度學習模型進行訓練，以取得預測值並繪製接收者操作特徵曲線，以決定本發明所決定出影像區塊是否具有癌症之閾值，藉以有效增進胰臟癌之辨識率。

- 醫療影像分析系統及其方法/MEDICAL IMAGE ANALYZING SYSTEM AND METHOD THEREOF

I767439

(19)中華民國智慧財產局

(12)發明說明書公告本　(11)證書號數：TW I767439 B

(45)公告日：中華民國 111 (2022) 年 06 月 11 日

(21)申請案號：109143455　(22)申請日：中華民國 109 (2020) 年 12 月 09 日

(51)Int. Cl.：

G06K9/62	*(2006.01)*	*G06K9/46*	*(2006.01)*
G06T7/00	*(2017.01)*	*G06T7/11*	*(2017.01)*
G06N3/08	*(2006.01)*	*G16H30/40*	*(2018.01)*

(30)優先權：2020/01/03　美國　62/956,687

(71)申請人：國立臺灣大學 (中華民國) NATIONAL TAIWAN UNIVERSITY (TW)
臺北市大安區羅斯福路 4 段 1 號

(72)發明人：廖偉智 LIAO, WEI CHIH (TW)；王偉仲 WANG, WEI CHUNG (TW)；劉高郎 LIU, KAO LANG (TW)；陳柏廷 CHEN, PO TING (TW)；張大衛 CHENG, DAWEI (TW)

(74)代理人：陳孚竹；張家彬

(56)參考文獻：

TW	201839634A	TW	201935403A
CN	109003260A	US	2012/0165217A1

審查人員：陳奕昌

申請專利範圍項數：28 項　圖式數：6　共 46 頁

(54)名稱

醫療影像分析系統及其方法

(57)摘要

本發明提供一種醫療影像分析系統及其方法，主要針對標記臟器具有癌症部位之分割標籤之處理影像進行擷取，以產生複數個影像區塊，並以複數個影像區塊進行特徵分析及對模型進行訓練，以取得預測值並繪製接收者操作特徵曲線，以決定出影像區塊是否具有癌症之閾值，藉以有效增進如胰臟癌之辨識率。

- 醫療影像分析系統及其方法/MEDICAL IMAGE ANALYZING SYSTEM AND METHOD THEREOF

I776716

(19)中華民國智慧財產局

(12)發明說明書公告本 **(11)證書號數：TW I776716 B**

(45)公告日： 中華民國 111 (2022) 年 09 月 01 日

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(21)申請案號：110139306　　(22)申請日： 中華民國 110 (2021) 年 10 月 22 日

(51)Int. Cl.：　*G16H30/40 (2018.01)*　*G06N3/08 (2006.01)*

(30)優先權：2020/10/23　美國　63/104,568

(71)申請人：國立臺灣大學 (中華民國) NATIONAL TAIWAN UNIVERSITY (TW)
臺北市大安區羅斯福路 4 段 1 號

(72)發明人：王偉仲 WANG, WEI CHUNG (TW)；廖偉智 LIAO, WEI CHIH (TW)；劉高郎 LIU, KAO LANG (TW)；陳柏廷 CHEN, PO TING (TW)；王柏川 WANG, PO CHUAN (TW)；吳亭慧 WU, TING HUI (TW)

(74)代理人：張家彬

(56)參考文獻：

期刊 Liu, Kao-Lang, et al. "Deep learning to distinguish pancreatic cancer tissue from non-cancerous pancreatic tissue: a retrospective study with cross-racial external validation." The Lancet Digital Health 2.6 (2020) Elsevier June 2020 pages e303-e313

審查人員：劉建宏

申請專利範圍項數：24 項　圖式數：2　共 25 頁

---

(54)名稱

醫療影像分析系統及方法

(57)摘要

本發明提供一種醫療影像分析系統及方法，主要將至少一病患影像輸入第一神經網路模組之第一模型，以得到具有確定該病患影像之臟器及腫瘤位置與範圍之結果，再將該結果分別輸入第二神經網路模組之複數第二模型，以得到對應各該複數第二模型之複數預測值以及該複數預測值中預測有癌症之模型數，最後根據該預測有癌症之模型數及一數量閾值，輸出一判斷結果。本發明可自動化第一模型及第二模型之間的流程，並有效增進胰臟癌之辨識率。

- 醫療影像分析系統及其方法/MEDICAL IMAGE ANALYZING SYSTEM FOR IDENTIFYING PANCREATIC CANCER USING MACHINE LEARNING AND METHOD THEREOF

US012106473B2

(12) **United States Patent**
Wang et al.

(10) **Patent No.: US 12,106,473 B2**
(45) **Date of Patent: *Oct. 1, 2024**

(54) **MEDICAL IMAGE ANALYZING SYSTEM FOR IDENTIFYING PANCREATIC CANCER USING MACHINE LEARNING AND METHOD THEREOF**

(71) Applicant: **National Taiwan University**, Taipei (TW)

(72) Inventors: **Wei-Chung Wang**, Taipei (TW); **Wei-Chih Liao**, Taipei (TW); **Kao-Lang Liu**, Taipei (TW); **Po-Ting Chen**, Taipei (TW); **Po-Chuan Wang**, Taipei (TW); **Da-Wei Chang**, Taipei (TW)

(73) Assignee: **National Taiwan University**, Taipei (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 447 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **17/507,949**

(22) Filed: **Oct. 22, 2021**

(65) **Prior Publication Data**

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Related U.S. Application Data

(60) Provisional application No. 63/104,568, filed on Oct. 23, 2020.

(51) **Int. Cl.**
G06T 7/00 (2017.01)
A61B 5/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***G06T 7/0012*** (2013.01); ***A61B 5/425*** (2013.01); ***A61B 5/4887*** (2013.01); ***A61B 5/726*** (2013.01);

(58) **Field of Classification Search**
CPC G06T 7/0012; G06T 7/73; G06T 7/70; G06T 2207/20016; G06T 2207/20064;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,803,987 B2 * 10/2020 Bériault G06N 3/045
11,583,698 B2 * 2/2023 Yin A61N 5/103

OTHER PUBLICATIONS

P. Wang, Automated Pancreas Segmentation Using Multi-institutional Collaborative Deep Learning, Sep. 2020 (Year: 2020).*
(Continued)

Primary Examiner — Stephen S Hong
Assistant Examiner — Carl E Barnes, Jr.
(74) *Attorney, Agent, or Firm* — Amin, Turocy & Watson, LLP

(57) **ABSTRACT**

A medical image analyzing system and a medical image analyzing method are provided and include inputting at least one patient image into a first model of a neural network module to obtain a result having determined positions and ranges of an organ and a tumor of the patient image; inputting the result into a second model of a first analysis module and a third model of a second analysis module, respectively, to obtain at least one first prediction value and at least one second prediction value corresponding to the patient image; and outputting a determined result based on the first prediction value and the second prediction value. Further, processes between the first model, the second model and the third model can be automated, thereby improving identification rate of pancreatic cancer.

- 醫療影像分析系統及其方法/MEDICAL IMAGE ANALYZING SYSTEM AND METHOD THEREOF

US012067717B2

(12) **United States Patent**
Wang et al.

(10) **Patent No.: US 12,067,717 B2**
(45) **Date of Patent: Aug. 20, 2024**

(54) **MEDICAL IMAGE ANALYZING SYSTEM AND METHOD THEREOF**

(71) Applicant: **National Taiwan University**, Taipei (TW)

(72) Inventors: **Wei-Chung Wang**, Taipei (TW); **Wei-Chih Liao**, Taipei (TW); **Kao-Lang Liu**, Taipei (TW); **Po-Ting Chen**, Taipei (TW); **Po-Chuan Wang**, Taipei (TW); **Ting-Hui Wu**, Taipei (TW)

(73) Assignee: **National Taiwan University**, Taipei (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 403 days.

(21) Appl. No.: **17/507,948**

(22) Filed: **Oct. 22, 2021**

(65) **Prior Publication Data**

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Related U.S. Application Data

(60) Provisional application No. 63/104,568, filed on Oct. 23, 2020.

(51) **Int. Cl.**
G06T 7/00 (2017.01)
A61B 5/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***G06T 7/0012*** (2013.01); ***A61B 5/425*** (2013.01); ***A61B 5/4887*** (2013.01); ***A61B 5/726*** (2013.01); ***A61B 5/7267*** (2013.01); ***G06N 3/08*** (2013.01); ***G06T 7/70*** (2017.01); ***G06T 7/73*** (2017.01); ***G06V 10/774*** (2022.01); ***G06V 10/776*** (2022.01); ***G06V 10/82*** (2022.01); ***G16H 30/20*** (2018.01); *G06T 2207/20016* (2013.01); *G06T 2207/20064* (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC G06T 7/00; G06T 7/0012; G06T 7/70; G06T 7/73; G16H 30/20; G06N 3/08; G06V 10/82; G06V 10/774; G06V 10/776; A61B 5/425; A61B 5/726; A61B 5/4887; A61B 5/7267
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,803,987	B2 *	10/2020	Bériault	G06N 3/045
11,583,698	B2 *	2/2023	Yin	A61N 5/103

* cited by examiner

Primary Examiner — Tuan H Nguyen
(74) *Attorney, Agent, or Firm* — Amin, Turocy & Watson, LLP

(57) **ABSTRACT**

A medical image analyzing system and a medical image analyzing method are provided and include inputting at least one patient image into a first model of a first neural network module to obtain a result having determined positions and ranges of an organ and a tumor of the patient image; inputting the result into a plurality of second models of a second neural network module, respectively, to obtain a plurality of prediction values corresponding to each of the plurality of second models and a model number predicting having cancer in the plurality of prediction values; and outputting a determined result based on the model number predicting having cancer and a number threshold value. Further, processes between the first model and the second models can be automated, thereby improving identification rate of pancreatic cancer.

22 Claims, 2 Drawing Sheets

- 醫療影像分析系統及其方法/MEDICAL IMAGE ANALYZING SYSTEM AND METHOD THEREOF

US011684333B2

(12) **United States Patent**
Liao et al.

(10) **Patent No.: US 11,684,333 B2**
(45) **Date of Patent: *Jun. 27, 2023**

(54) **MEDICAL IMAGE ANALYZING SYSTEM AND METHOD THEREOF**

(71) Applicant: **National Taiwan University**, Taipei (TW)

(72) Inventors: **Wei-Chih Liao**, Taipei (TW); **Wei-Chung Wang**, Taipei (TW); **Kao-Lang Liu**, Taipei (TW); **Po-Ting Chen**, Taipei (TW); **Da-Wei Chang**, Taipei (TW)

(73) Assignee: **National Taiwan University**, Taipei (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 248 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **17/137,647**

(22) Filed: **Dec. 30, 2020**

(65) **Prior Publication Data**

US 2021/0204898 A1 Jul. 8, 2021

Related U.S. Application Data

(60) Provisional application No. 62/956,687, filed on Jan. 3, 2020.

(30) **Foreign Application Priority Data**

Dec. 9, 2020 (TW) 109143455

(51) **Int. Cl.**
A61B 6/00 (2006.01)
G06T 7/11 (2017.01)
(Continued)

(52) **U.S. Cl.**
CPC ***A61B 6/5217*** (2013.01); ***G06T 7/0012*** (2013.01); ***G06T 7/11*** (2017.01);
(Continued)

(58) **Field of Classification Search**
CPC A61B 6/5217; A61B 5/055; A61B 6/032; A61B 5/0013; A61B 5/4887;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,346,728 B2 * 7/2019 Fu G06V 30/194
10,489,908 B2 * 11/2019 Kiraly G06T 7/0012
(Continued)

FOREIGN PATENT DOCUMENTS

CN	109003260	12/2018
TW	201839634	11/2018
TW	201935403	9/2019

OTHER PUBLICATIONS

Hang Zhou,Targeted attack and security enhancement on texture synthesis based steganography, Journal of Visual Communication and Image Representation,vol. 54, (Year: 2018).*
(Continued)

Primary Examiner — Boniface Ngathi N
Assistant Examiner — Zainab Mohammed Aldarraji
(74) *Attorney, Agent, or Firm* — Amin, Turocy & Watson, LLP

(57) **ABSTRACT**

Provided is a medical image analyzing system and a method thereof, which includes: acquiring a processed image having a segmentation label corresponding to a cancerous part of an organ (if present), generating a plurality of image patches therefrom, performing feature analysis on the image patches and model training to obtain prediction values, drawing a receiver operating characteristic curve using the prediction values, and determining a threshold with which determines
(Continued)

- 醫療影像分析系統及其方法/MEDICAL IMAGE ANALYZING SYSTEM AND METHOD THEREOF

US011424021B2

(12) **United States Patent**
Liao et al.

(10) **Patent No.: US 11,424,021 B2**
(45) **Date of Patent: Aug. 23, 2022**

(54) **MEDICAL IMAGE ANALYZING SYSTEM AND METHOD THEREOF**

(71) Applicant: **National Taiwan University**, Taipei (TW)

(72) Inventors: **Wei-Chih Liao**, Taipei (TW); **Wei-Chung Wang**, Taipei (TW); **Kao-Lang Liu**, Taipei (TW); **Po-Ting Chen**, Taipei (TW); **Ting-Hui Wu**, Taipei (TW); **Holger Roth**, Taipei (TW)

(73) Assignee: **National Taiwan University**, Taipei (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 195 days.

(21) Appl. No.: **16/868,742**

(22) Filed: **May 7, 2020**

(65) **Prior Publication Data**

US 2020/0357506 A1 Nov. 12, 2020

Related U.S. Application Data

(60) Provisional application No. 62/845,922, filed on May 10, 2019.

(30) **Foreign Application Priority Data**

Apr. 21, 2020 (TW) 109113320

(51) **Int. Cl.**
G06K 9/00 (2022.01)
G16H 30/40 (2018.01)
(Continued)

(52) **U.S. Cl.**
CPC ***G16H 30/40*** (2018.01); ***G06N 3/08*** (2013.01); ***G06T 7/0012*** (2013.01); ***G06T 7/11*** (2017.01);
(Continued)

(58) **Field of Classification Search**
CPC G16H 30/40
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,486,142 B2	11/2016	Hielscher et al.		
2014/0105471 A1*	4/2014	Brown	....	G06T 7/11 382/128

(Continued)

FOREIGN PATENT DOCUMENTS

CN	107767362	3/2018
CN	109584252	4/2019

(Continued)

OTHER PUBLICATIONS

Liao, et al. "Differentiation Between Pancreatic Cancer and Normal Pancreas on Computed Tomotraphy with Artificial Intelligence", A. Gastroenterology supplement to DDW, vol. 156, Issue 6, Supplement 1, p. S-59, https://www.gastrojournal.org/issue/S0016-5085(19)X6001-1?page=3.

(Continued)

Primary Examiner — Hadi Akhavannik
(74) *Attorney, Agent, or Firm* — Amin, Turocy & Watson, LLP

(57) **ABSTRACT**

Provided are a medical image analyzing system and a method thereof, which mainly crop a plurality of image patches from a processed image including a segmentation label corresponding to a location of an organ, train a deep learning model with the image patches to obtain prediction values, and plot a receiver operating characteristic curve to determine a threshold which determines whether the image patches are cancerous, thereby effectively improving the detection rate of cancer.

18 Claims, 10 Drawing Sheets

【19】中華民國　　　　　　　　【12】專利公報　　(B)

【11】證書號數：I918507
【45】公告日：中華民國 115 (2026) 年 03 月 11 日
【51】Int. Cl.：　　*G16H30/40 (2018.01)*　　*G16H50/20 (2018.01)*
G06T7/00 (2017.01)　　*G06T5/50 (2006.01)*
發明　　全 7 頁

【54】名　稱：進階影像分析系統、方法及其電腦可讀媒介
【21】申請案號：114118895　　【22】申請日：中華民國 114 (2025) 年 05 月 20 日
【11】公開編號：202609795　　【43】公開日期：中華民國 115 (2026) 年 03 月 01 日
【30】優 先 權：2024/05/20　　美國　　63/649,801
【72】發 明 人：王偉仲 (TW) WANG, WEI-CHUNG；廖偉智 (TW) LIAO, WEI-CHIH；陳柏廷 (TW) CHEN, PO-TING；張大衛 (TW) CHANG, DA-WEI；陳彥嘉 (TW) CHEN, YEN-JIA；葉彥辰 (TW) YEH, YAN-CHEN
【71】申 請 人：國立臺灣大學　　NATIONAL TAIWAN UNIVERSITY
臺北市大安區羅斯福路 4 段 1 號
國立臺灣大學醫學院附設醫院　　NATIONAL TAIWAN UNIVERSITY HOSPITAL
臺北市中正區中山南路 7 號
【74】代 理 人：潘柏均
【56】參考文獻：

CN	115937620A	CN	116128855A
CN	118016228A	US	2021/0217167A1
US	2021/0407082A1	US	2023/0090906A1

審查人員：施易眆

【57】申請專利範圍

1. 一種進階影像分析系統，係包括：
影像預處理模組，係用於電腦斷層影像中腹部和指定器官之定位，將該電腦斷層影像之立體像素的亨氏單位大於一第一設定值的部分視為非空氣區域，以據之確認主體區域，以及將該電腦斷層影像之軸向切片中所佔最大成分小於一第二設定值者刪除，以定義出軀幹位置，之後，計算該軀幹位置的每一切片的內部空氣比例，以確定出腹部位置，俾將該腹部位置的影像透過插值法重新取樣並進行視窗處理和標準化，以成為預處理影像；
影像判斷模組，係連結該影像預處理模組且具有一分析單元，該分析單元用於從該預處理影像提取切片特徵以產生對應的標記，透過分析各切片對應的該標記以及經訓練後的分類標記以判斷出該指定器官是否有疾病，以及將該切片本身、前一切片和後一切片的總注意力值最大者作為最重要峰值，以令對應該最重要峰值之該切片本身、該前一切片和該後一切片成為標示該指定器官為有疾病之關鍵影像切片；以及
成像模組，係連結該影像判斷模組，用以顯示該關鍵影像切片，以提供醫療人員查看。
2. 如請求項 1 所述之進階影像分析系統，其中，該分析單元復包括：
可訓練的二維模型，係用於對該預處理影像提取切片特徵以成為該標記；以及
可訓練的轉換器，係用於將各該切片對應的該標記與該分類標記結合，經注意力機制分析後，從各該切片對應的該標記的注意力值中找出該最重要峰值，藉以得到該關鍵影像切片。

【19】中華民國　　　【12】發明公開公報　(A)

【11】公開編號：202609795　　　　　　申請實體審查：有

【43】公開日：中華民國 115 (2026) 年 03 月 01 日

【51】Int. Cl.：　*G16H30/40　(2018.01)*　　*G16H50/20　(2018.01)*

G06T7/00　(2017.01)　　*G06T5/50　(2006.01)*

【54】發明名稱：進階影像分析系統、方法及其電腦可讀媒介

【21】申請案號：114118895　　【22】申請日：中華民國 114 (2025) 年 05 月 20 日

【30】優 先 權：2024/05/20　　美國　　63/649,801

【72】發 明 人：王偉仲 (TW) WANG, WEI-CHUNG；廖偉智 (TW) LIAO, WEI-CHIH；陳柏廷 (TW) CHEN, PO-TING；張大衛 (TW) CHANG, DA-WEI；陳彥嘉 (TW) CHEN, YEN-JIA；葉彥辰 (TW) YEH, YAN-CHEN

【71】申 請 人：國立臺灣大學　NATIONAL TAIWAN UNIVERSITY

臺北市大安區羅斯福路 4 段 1 號

國立臺灣大學醫學院附設醫院　NATIONAL TAIWAN UNIVERSITY HOSPITAL

臺北市中正區中山南路 7 號

【74】代 理 人：潘柏均

【57】發明摘要：

本發明揭露一種進階影像分析系統及其方法，透過卷積神經網路(CNN)搭配轉換器(transformer)，建構出一深度學習模型來進行預測分析，將電腦斷層影像經過預處理得到預處理影像，該預處理影像由分析單元進行分析以判斷器官是否有疾病，且讓具有最重要峰值之切片本身、其前一切片和其後一切片成為關鍵影像切片，以標記和顯示出對應影像以供醫護人員查看，據上，本發明能以無使用顯影劑之電腦斷層影像來檢測腹部中器官是否有疾病。本發明復提供一種電腦可讀媒介，係用於執行本發明之方法。

指定代表圖：

- 2485 -

2. 國內外獲獎證明及說明

年度	獎項名稱	頒發單位	受獎單位	獎項來源
2025	國家藥物科技研究發展獎-金質獎	衛福部與經濟部	仲智數位健康股份有限公司	國內獎項
2024	國家生技醫療品質獎(SNQ)-銀獎	國家生技醫療產業策進會(生策會)	仲智數位健康股份有限公司	國內獎項
2024	台北生技獎-金獎	臺北市政府	仲智數位健康股份有限公司	國內獎項
2024	第 21 屆國家新創獎-新創精進獎	國家生技醫療產業策進會(生策會)	國立臺灣大學醫學院附設醫院	國內獎項
2023	Alexander R. Margulis Award	北美放射醫學會 (RSNA) (放射醫學界極具份量的年度科學獎，台灣首獲此殊榮)	國立臺灣大學	國外獎項
2023	第 20 屆國家新創獎 - 新創精進獎	國家生技醫療產業策進會(生策會)	國立臺灣大學醫學院附設醫院	國內獎項
2022	第 19 屆國家新創獎 - 新創精進獎	國家生技醫療產業策進會(生策會)	國立臺灣大學醫學院附設醫院	國內獎項
2022	突破性醫材認定 (Breakthrough Device Designation)	美國食品藥物管理局 (FDA)	仲智數位健康股份有限公司	國外獎項
2021	第 18 屆國家新創獎 - 學研新創獎	國家生技醫療產業策進會(生策會)	國立臺灣大學醫學院附設醫院	國內獎項

- 國家藥物科技研究發展獎-**金質獎**


NPRDA

2025國家藥物科技研究發展獎

藥物領航 躍升國際

仲智數位健康股份有限公司

人工智慧胰臟癌輔助偵測系統

-PANCREASaver 助胰見

參加 114 年度國家藥物科技研究發展獎

榮獲「醫療器材類金質獎」，特頒此狀，以資鼓勵

中華民國114年11月18日

FDA 衛生福利部 食品藥物管理署

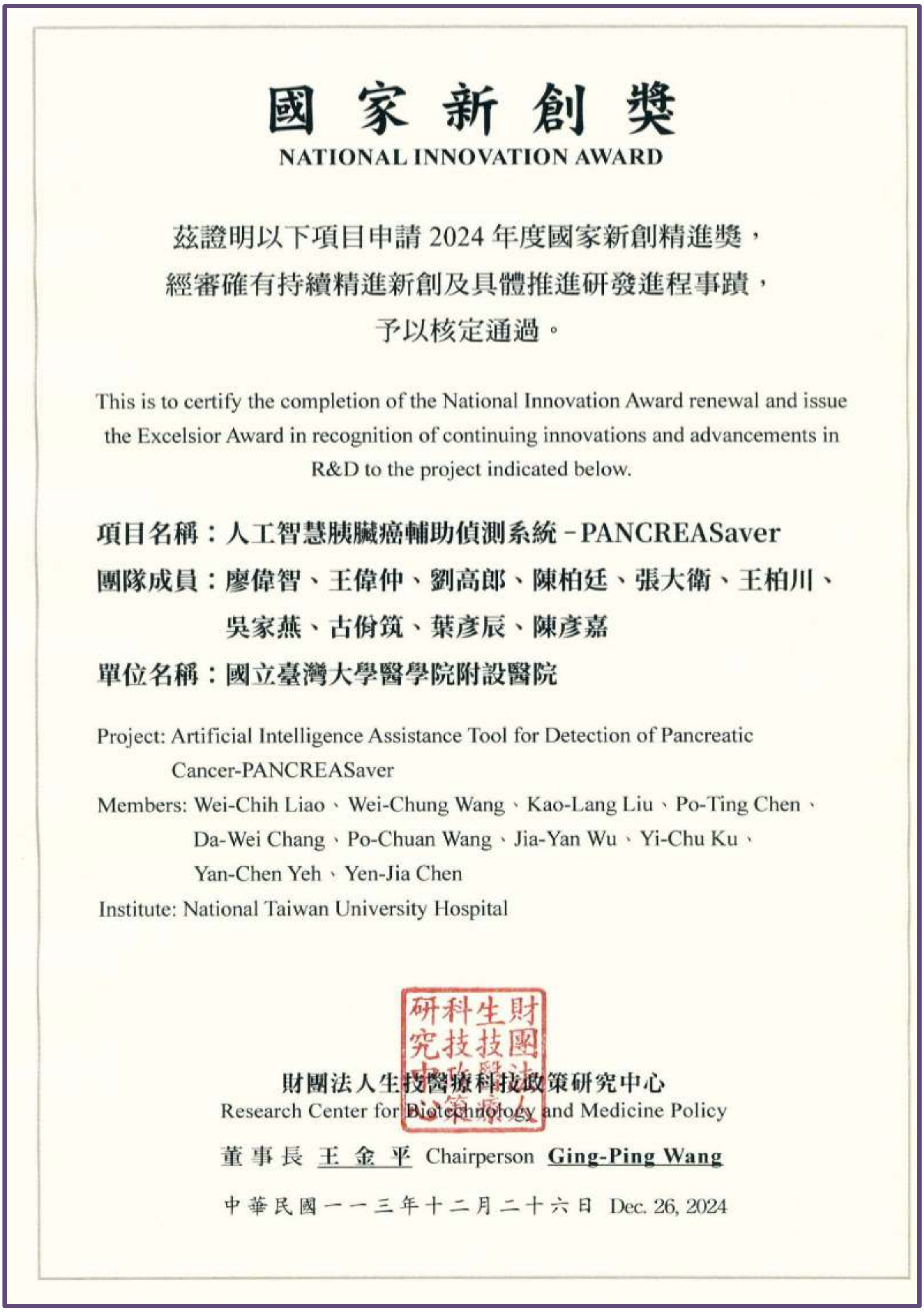
經濟部 產業發展署

國家生技醫療品質獎 (SNQ)-銀獎

- 台北生技獎-金獎


TAIPEI
臺北
2024臺北生技獎
跨域卓越獎
金 獎
仲智數位健康股份有限公司
臺北市長 蔣萬安 敬贈
中華民國113年9月24日
2024 Taipei Biotech Awards
Interdisciplinary Excellence Gold Award
PanCAD.ai CO., Ltd.
Mayor of Taipei
September 24, 2024

- 第 21 屆國家新創獎-新創精進獎



國家新創獎
NATIONAL INNOVATION AWARD

茲證明以下項目申請 2024 年度國家新創精進獎，
經審確有持續精進新創及具體推進研發進程事蹟，
予以核定通過。

This is to certify the completion of the National Innovation Award renewal and issue the Excelsior Award in recognition of continuing innovations and advancements in R&D to the project indicated below.

項目名稱：人工智慧胰臟癌輔助偵測系統－PANCREASaver
團隊成員：廖偉智、王偉仲、劉高郎、陳柏廷、張大衛、王柏川、吳家燕、古佾筑、葉彥辰、陳彥嘉
單位名稱：國立臺灣大學醫學院附設醫院

Project: Artificial Intelligence Assistance Tool for Detection of Pancreatic Cancer-PANCREASaver
Members: Wei-Chih Liao、Wei-Chung Wang、Kao-Lang Liu、Po-Ting Chen、Da-Wei Chang、Po-Chuan Wang、Jia-Yan Wu、Yi-Chu Ku、Yan-Chen Yeh、Yen-Jia Chen
Institute: National Taiwan University Hospital

財團法人生技醫療科技政策研究中心
Research Center for Biotechnology and Medicine Policy

董事長 王金平 Chairperson Ging-Ping Wang

中華民國一一三年十二月二十六日 Dec. 26, 2024

- **Alexander R. Margulis Award**

RSNA®

Radiological Society of North America

Presents this

2023 Alexander R. Margulis Award for Scientific Excellence

To

"Pancreatic Cancer Detection on CT Scans with Deep Learning: A Nationwide Population-based Study"

By

Po-Ting Chen, MD
Tinghui Wu, MS
Pochuan Wang, MS
Dawei Chang, MS
Kao-Lang, Liu, MD
Ming-Shiang Wu, MD, PhD
Holger R. Roth, PhD
Po-Chang Lee, MD
Wei-Chih Liao, MD, PhD
Weichung Wang, MD

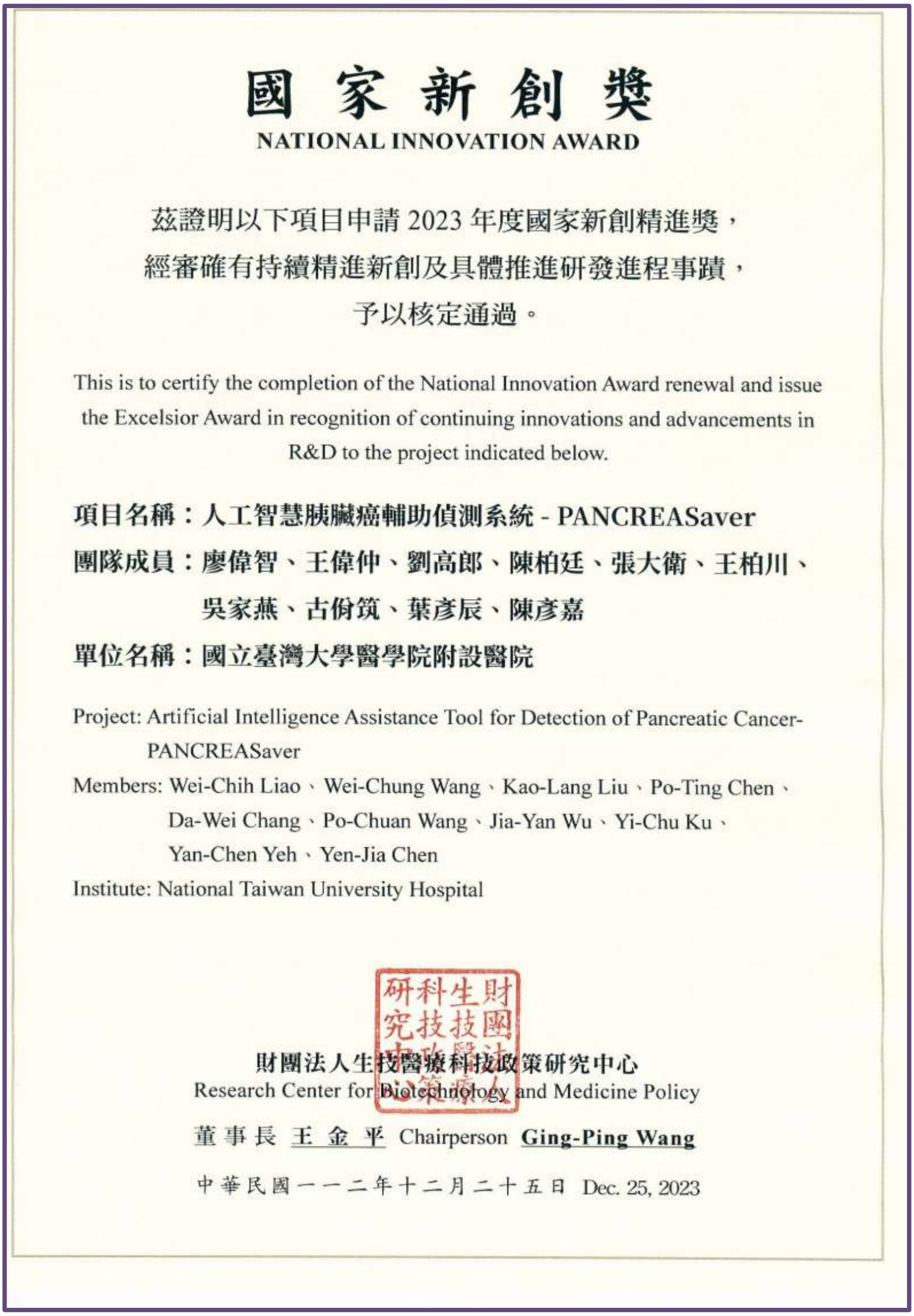
Radiology 2023; 306:172182

CHICAGO, ILLINOIS – NOVEMBER 27, 2023

Matthew A. Mauro, MD
President

Umar Mahmood, MD, PhD
Secretary-Treasurer

- 第 20 屆國家新創獎 - 新創精進獎



國 家 新 創 獎

NATIONAL INNOVATION AWARD

茲證明以下項目申請 2023 年度國家新創精進獎，
經審確有持續精進新創及具體推進研發進程事蹟，
予以核定通過。

This is to certify the completion of the National Innovation Award renewal and issue the Excelsior Award in recognition of continuing innovations and advancements in R&D to the project indicated below.

項目名稱：人工智慧胰臟癌輔助偵測系統 - PANCREASaver

團隊成員：廖偉智、王偉仲、劉高郎、陳柏廷、張大衛、王柏川、吳家燕、古佾筑、葉彥辰、陳彥嘉

單位名稱：國立臺灣大學醫學院附設醫院

Project: Artificial Intelligence Assistance Tool for Detection of Pancreatic Cancer-PANCREASaver

Members: Wei-Chih Liao、Wei-Chung Wang、Kao-Lang Liu、Po-Ting Chen、Da-Wei Chang、Po-Chuan Wang、Jia-Yan Wu、Yi-Chu Ku、Yan-Chen Yeh、Yen-Jia Chen

Institute: National Taiwan University Hospital

財團法人生技醫療科技政策研究中心

Research Center for Biotechnology and Medicine Policy

董事長 王金平 Chairperson **Ging-Ping Wang**

中華民國一一二年十二月二十五日 Dec. 25, 2023

- 第 19 屆國家新創獎 - 新創精進獎

國家新創獎
NATIONAL INNOVATION AWARD

茲證明以下項目申請 2022 年度國家新創精進獎，
經審確有持續精進新創及具體推進研發進程事蹟，
予以核定通過。

This is to certify the completion of the National Innovation Award renewal and issue the Excelsior Award in recognition of continuing innovations and advancements in R&D to the project indicated below.

項目名稱：人工智慧胰臟癌輔助偵測系統 – PANCREASaver
團隊成員：廖偉智、王偉仲、劉高郎、陳柏廷、吳亭慧、張大衛、
王柏川、吳家燕、古佾筑、葉彥辰、陳彥嘉、謝侑倫
單位名稱：國立臺灣大學醫學院附設醫院

Project: Artificial Intelligence Assistance Tool for Detection of Pancreatic Cancer - PANCREASaver
Members: Wei-Chih Liao, Wei-Chung Wang, Kao-Lang Liu, Po-Ting Chen, Ting-Hui Wu, Da-Wei Chang, Po-Chuan Wang, Jia-Yan Wu, Yi-Chu Ku, Yan-Chen Yeh, Yen-Jia Chen, Yu-Lun Hsieh
Institute: National Taiwan University Hospital

財團法人生技醫療科技政策研究中心
Research Center for Biotechnology and Medicine Policy

董事長 王金平 Chairperson **Ging-Ping Wang**

中華民國一一一年十二月二十三日 Dec. 23, 2022

- 突破性醫材認定 (Breakthrough Device Designation)

U.S. FOOD & DRUG ADMINISTRATION

November 30, 2022

[redacted]

Re: Q[redacted]/S001
Trade/Device Name: PANCREASaver
Received: October 11, 2022

Dear Wei-Chih Liao:

The Center for Devices and Radiological Health (CDRH) of the Food and Drug Administration (FDA) has received the above submission requesting designation as a Breakthrough Device. The proposed indications for use includes "PANCREASaver is a computer-assisted detection and diagnosis (CADe/x) software device for use in healthcare facilities and/or hospitals to assist qualified physicians in the detection and diagnosis of pancreatic ductal adenocarcinoma (PDAC) using venous phase contrast-enhanced CT images obtained from adult patients. [redacted] By providing information that may be useful in the detection and diagnosis of PDAC during image interpretation, PANCREASaver provides complementary information in assisting qualified physician interpretation.

Please note: Patient management decisions should not be made solely on the results of the PANCREASaver analysis.." We are pleased to inform you that your device and proposed indication for use meet the criteria and have been granted designation as a Breakthrough Device. Please refer to the FDA guidance document entitled "Breakthrough Devices Program", for more information regarding the program, available at https://www.fda.gov/media/108135/download.

We recommend you review the FDA guidance document for the Breakthrough Devices Program referenced above for the available mechanisms for obtaining feedback from the Agency on device development for designated breakthrough devices. When submitting any new requests, please reference Q[redacted]/S001. Any new submission should be provided as an eCopy, it should include the FDA reference number for this submission, and should be submitted to the following address:

U.S. Food & Drug Administration
10903 New Hampshire Avenue
Silver Spring, MD 20993
www.fda.gov

U.S. Food and Drug Administration
Center for Devices and Radiological Health
IDE Document Control Center - WO66-G609
10903 New Hampshire Avenue
Silver Spring, MD 20993-0002

[redacted] Please also note that any Premarket Approval (PMA) application must be signed by an authorized representative residing or maintaining a place of business in the United States (21 CFR 814.20).

[redacted]

If you have any questions, please contact Wenbo Li at Wenbo.Li@fda.hhs.gov.

Sincerely,

Laurel Burk, Ph.D.
Director
DHT8B: Division of Radiological Imaging
Devices and Electronic Products
OHT8: Office of Radiological Health
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

- 第 18 屆國家新創獎 - 學研新創獎

3. 國內外期刊、論文發表

年	文獻	期刊	類別
2023	Pancreatic Cancer Detection on CT with Deep Learning: Nationwide Study	Radiology	頂尖期刊
2023	2D/3D Radiomic Analysis, Nationwide Real-world Dataset	BMC Cancer	期刊
2021	Radiomic Features at CT Distinguish Pancreatic Cancer	Radiology: Imaging Cancer	期刊
2021	Applications of AI in Pancreatic and Biliary Diseases	J Gastroenterol Hepatol	期刊
2020	Deep Learning to Distinguish Pancreatic Cancer Tissue	Lancet Digital Health	頂尖期刊

Radiology

Pancreatic Cancer Detection on CT Scans with Deep Learning: A Nationwide Population-based Study

*Po-Ting Chen, MD** • *Tinghui Wu, MS** • *Pochuan Wang, MS* • *Dawei Chang, MS* • *Kao-Lang Liu, MD* • *Ming-Shiang Wu, MD, PhD* • *Holger R. Roth, PhD* • *Po-Chang Lee, MD* • *Wei-Chih Liao, MD, PhD*** • *Weichung Wang, PhD***

From the Department of Medical Imaging (P.T.C., K.L.L.) and Division of Gastroenterology and Hepatology, Department of Internal Medicine (M.S.W., W.C.L.), National Taiwan University Hospital, National Taiwan University College of Medicine, Taipei, Taiwan; Institute of Applied Mathematical Sciences (T.W., D.C., W.W.) and Departments of Computer Science and Information Engineering (P.W.) and Internal Medicine, College of Medicine (M.S.W., W.C.L.), National Taiwan University, No. 1, Section 4, Roosevelt Rd, Taipei 10617, Taiwan; Department of Medical Imaging, National Taiwan University Cancer Center, Taipei, Taiwan (K.L.L.); NVIDIA, Bethesda, Md (H.R.R.); and National Health Insurance Administration, Ministry of Health and Welfare, Taipei, Taiwan (P.C.L.). Received March 5, 2022; revision requested April 22; revision received July 15; accepted August 1. **Address correspondence to** W.W. (email: wwang@ntu.edu.tw).

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* P.T.C. and T.W. contributed equally to this work.

** W.C.L. and W.W. are co-senior authors.

Conflicts of interest are listed at the end of this article.

See also the editorial by Aisen and Rodrigues in this issue.

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Background: Approximately 40% of pancreatic tumors smaller than 2 cm are missed at abdominal CT.

Purpose: To develop and to validate a deep learning (DL)–based tool able to detect pancreatic cancer at CT.

Materials and Methods: Retrospectively collected contrast-enhanced CT studies in patients diagnosed with pancreatic cancer between January 2006 and July 2018 were compared with CT studies of individuals with a normal pancreas (control group) obtained between January 2004 and December 2019. An end-to-end tool comprising a segmentation convolutional neural network (CNN) and a classifier ensembling five CNNs was developed and validated in the internal test set and a nationwide real-world validation set. The sensitivities of the computer-aided detection (CAD) tool and radiologist interpretation were compared using the McNemar test.

Results: A total of 546 patients with pancreatic cancer (mean age, 65 years ± 12 [SD], 297 men) and 733 control subjects were randomly divided into training, validation, and test sets. In the internal test set, the DL tool achieved 89.9% (98 of 109; 95% CI: 82.7, 94.9) sensitivity and 95.9% (141 of 147; 95% CI: 91.3, 98.5) specificity (area under the receiver operating characteristic curve [AUC], 0.96; 95% CI: 0.94, 0.99), without a significant difference (P = .11) in sensitivity compared with the original radiologist report (96.1% [98 of 102]; 95% CI: 90.3, 98.9). In a test set of 1473 real-world CT studies (669 malignant, 804 control) from institutions throughout Taiwan, the DL tool distinguished between CT malignant and control studies with 89.7% (600 of 669; 95% CI: 87.1, 91.9) sensitivity and 92.8% specificity (746 of 804; 95% CI: 90.8, 94.5) (AUC, 0.95; 95% CI: 0.94, 0.96), with 74.7% (68 of 91; 95% CI: 64.5, 83.3) sensitivity for malignancies smaller than 2 cm.

Conclusion: The deep learning–based tool enabled accurate detection of pancreatic cancer on CT scans, with reasonable sensitivity for tumors smaller than 2 cm.

Online supplemental material is available for this article.

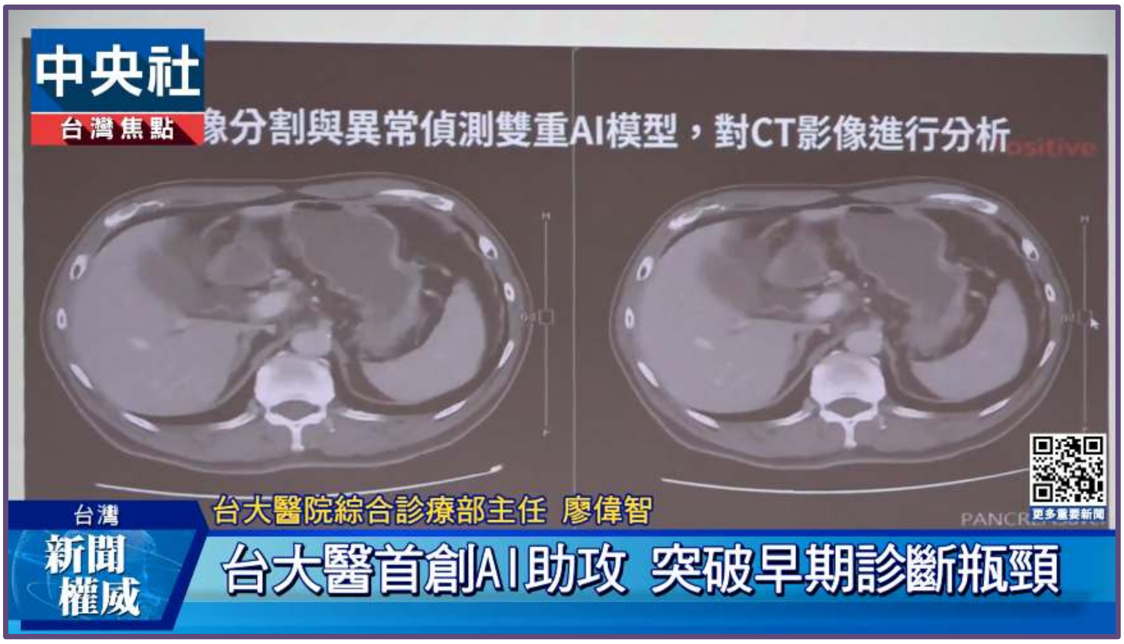
Pancreatic cancer (PC) has the lowest 5-year survival rate among cancers and is projected to become the second leading cause of cancer death in the United States by 2030 (1). Because the prognosis worsens precipitously once the tumor grows larger than 2 cm, early detection represents the most effective strategy to improve the dismal prognosis of PC (2). CT is the major imaging modality used to help detect PC, but its sensitivity for small tumors is modest, with approximately 40% of tumors smaller than 2 cm being missed (3). Furthermore, the diagnostic performance of CT is interpreter dependent and may be influenced by disparities in radiologist availability and expertise. Therefore, an effective tool to supplement radiologists in improving the sensitivity for PC detection is needed and constitutes a major unmet medical need.

Recent advances in deep learning (DL) have shown great promise in medical image analysis (4). In our previous single-center proof-of-concept study, we showed that a convolutional neural network (CNN) could accurately distinguish PC from noncancerous pancreas (5). However, in that study, segmentation of the pancreas (ie, identification of the pancreas) for analysis by the CNN was manually performed by radiologists. Segmentation of the pancreas is most challenging (6) because the pancreas borders multiple organs and structures and varies widely in shape and size, especially in patients with PC. However, a clinically applicable computer-aided detection (CAD) tool should enable segmentation and classification (ie, predicting presence or absence of PC), with minimal human annotation or labor.

PanCAD.ai

4. 醫學雜誌、其他媒體報導

	新聞報導標題	來源頻道	發佈時間	URL
1	早期診斷胰臟癌！ 台大醫啟用 AI 系統「助胰見」	TTV LIVE 台視直播	2026/5/4	https://www.youtube.com/watch?v=WjaYr4B2sVE
2	台大醫首創 AI 偵測 揪胰臟癌早期病灶	中央社影音新聞	2025/5/20	https://www.youtube.com/watch?v=GVtCDu0WSuI
3	台大醫研發全球首套 AI 系統 2 分鐘揪胰臟癌病灶	公視中晝新聞	2025/05/20	https://www.youtube.com/watch?v=WVe38gA7jEI
4	台大醫率全球之先 開發偵測胰臟癌利器 | 大愛新聞	大愛新聞 DaaiNews	2025/05/20	https://www.youtube.com/watch?v=5Qdw7U0KzRs
5	揪出小於 2 公分早期胰臟癌 台大首創 AI 診斷輔助系統	公視晚間新聞	2024/01/23	https://www.youtube.com/watch?v=CaeXd0_wilI


中央社
台灣焦點
象分割與異常偵測雙重AI模型，對CT影像進行分析
台大醫院綜合診療部主任 廖偉智
台灣
新聞
權威
台大醫首創AI助攻 突破早期診斷瓶頸


PANCREASaver
影像組學 x 機器學習
深度學習
有胰臟癌
無胰臟癌
胰臟/腫瘤位置
助胰見診斷胰臟癌
中畫新聞
Tiong-tàu sin-bûn
台大醫院研發全球首套AI系統 揪出胰臟癌


大愛
AI輔助醫療偵測 終結診斷盲區
以影像分割與異常偵測雙重AI模型，對CT影像進行分析
創新
2
快速
2
準確
9
午安新聞
"助胰見"
AI輔助診療系統 及早發現胰臟癌


公視
19:47:39
晚間新聞
PTS EVENING NEWS
台大推AI診斷輔助系統 辨識早期胰臟癌

	報導標題	日期	報導來源	URL
1	AI 讓胰臟癌及早被發現、救回一命！台大醫師廖偉智：3 種人應該及早篩檢	2025/10/30	50+(遠見天下事業群)	https://www.fiftyplus.com.tw/articles/33032
2	臺大醫院啟用 AI 系統「助胰見」 突破胰臟癌早期診斷瓶頸	2025/5/22	優活健康	https://www.uho.com.tw/article-66684.html
3	台大首創 AI 系統 「揪出早期胰臟癌」	2025/5/21	NOWnews 今日新聞	https://tw.news.yahoo.com/台大首創 ai 系統-揪出早期胰臟癌-053907207.html?guccounter=1&guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlLmNvbS8&guce_referrer_sig=AQAAAHRrlFVcVncbSKRAKT-RfTw2FUwlptT52ILdSFqXa7WqxxRd0j7HA47sXV4nMmVAEoA-pd3M-AQvvcOrDsJ3qEn01knhuVwy-uAkj3LgcyfAPY5DYV6XnGYs_AMKQzbh1NBNShzN_ZaeCR3eNKItKqG9LQQevFv3hEH_mrngL-HE
4	「台大 AI 助手」好給力 突破瓶頸！2 分鐘揪出早期胰臟癌	2025/5/21	CTWANT	https://www.ctwant.com/article/418278/
5	台大醫 AI 助手 2 分鐘揪出早期胰臟癌	2025/5/21	UDN 元氣網	https://health.udn.com/health/story/6015/8753581
6	全球首創！台大醫院 AI 系統揪出早期癌王 5 年存活率可達 8 成	2025/05/20	自由健康網	https://health.ltn.com.tw/article/breakingnews/5048613
7	台大醫院首套 AI 胰臟癌輔助診斷 2 分鐘揪早期病灶	2025/5/20	中央通訊社	https://www.cna.com.tw/news/ahel/202505200099.aspx
8	「胰臟癌」早期難發現！台大研發 AI 系統 2 分鐘偵測病兆	2025/5/20	HEHO	https://heho.com.tw/archives/356421
9	癌王剋星出世！ 台大醫院領先全球 開發首套 AI 早期診斷系統	2025/5/20	Rti 央廣新聞	https://www.rti.org.tw/news?uid=3&pid=141963
10	診斷極兇癌王有利器！台大 AI 系統及早揪出胰臟癌 存活率 10%變 80%	2025/5/20	健康 2.0	https://health.tvbs.com.tw/cancer/355056

	報導標題	日期	報導來源	URL
11	台大醫研發全球首套 AI 系統 2 分鐘揪胰臟癌病灶	2025/5/20	公視新聞網	https://news.pts.org.tw/article/752026
12	讓癌王無所遁形！台大開發 AI 診斷系統「助胰見」　精準揪出早期胰臟癌	2025/5/20	風傳媒	https://www.storm.mg/article/5376230
13	輝達 GTC 將發表｜台大 AI 輔助診斷胰臟癌領先全球　存活率大增至 80%	2025/05/20	知新聞	https://www.knews.com.tw/news/59BD131AA8827B2BF0DEF1663E130963
14	新創幫專欄｜早期偵測胰臟癌！台大廖偉智團隊「PANCREASaver 助胰見」取得 TFDA 將在台落地	2023/12/26	未來城市(天下)	https://futurecity.cw.com.tw/article/3326
15	臺大胰臟癌 AI 檢測研究-臺灣首度獲放射醫學國際大獎 Margulis Award 殊榮	2023/12/13	台灣大學	https://www.ntu.edu.tw/spotlight/2023/2221_20231213.html
16	科研新創助力健康照護 - AI 智慧輔助診斷　強化照護醫療	2023/11/10	僑新聞	https://ocacnews.net/article/354417
17	台大研發 AI 胰臟癌輔助偵測系統 獲美 FDA 突破醫材認定	2023/11/6	自由健康網	https://health.ltn.com.tw/article/breakingnews/4481249
18	特別企劃 / 胰臟癌 AI 有助於早期偵測	2022/10/15	好心肝會刊	https://www.liver.org.tw/journalView.php?cat=76&sid=1131
19	胰臟癌》「癌王」發現 + 治療都難　AI 輔助及早診斷大突破	2022/03/01	康健雜誌	https://www.commonhealth.com.tw/article/85883?utm_source=copyshare
20	癌王胰臟癌難發現　2 徵兆是重要線索　AI 輔助早期偵測	2022/2/21	康健雜誌	https://www.commonhealth.com.tw/article/85850
21	台大醫研發人工智慧揪早期胰臟癌 存活率提升逾十倍	2020/10/30	未來城市(天下)	https://futurecity.cw.com.tw/article/1743
22	AI-Based Method for Early Pancreatic Cancer Detection Receives Margulis Award		RSNA	https://www.rsna.org/annual-meeting/awards-recognition/2023-margulis

5. 國內外相關醫材證照

衛生福利部醫療器材許可證

衛部醫器製字第 007946 號

中文品名：“助胰見”人工智慧胰臟癌輔助偵測系統

英文品名：“PANCREASaver” Image Analytics Software

類　　別：第 P 類：放射學科學

醫療器材商名稱：仲智數位健康股份有限公司

規格或型號：PANCREASaver

以下空白

製造業者名稱：仲智數位健康股份有限公司委託商之器科技股份有限公司製造

製造業者地址：臺北市內湖區內湖路 1 段 516 號 5 樓

效能、用途或適應症：詳如核定之中文說明書

前項醫療器材經本部審核與醫療器材管理法之規定相符應發給許可證以資證明

衛生福利部部長

薛瑞元

衛生福利部印

發證日期 112 年 07 月 13 日

有效日期 117 年 07 月 13 日

核准展延至	年　月　日	年　月　日	年　月　日	年　月　日
文號				

MF 016572

2、 臨床報告文件及說明

使用國內全國性資料驗證並發表於《Radiology》

ORIGINAL RESEARCH • GASTROINTESTINAL IMAGING

Radiology

Pancreatic Cancer Detection on CT Scans with Deep Learning: A Nationwide Population-based Study

Po-Ting Chen, MD • Tinghui Wu, MS* • Pochuan Wang, MS • Dawei Chang, MS • Kao-Lang Liu, MD • Ming-Shiang Wu, MD, PhD • Holger R. Roth, PhD • Po-Chang Lee, MD • Wei-Chih Liao, MD, PhD** • Weichung Wang, PhD***

From the Department of Medical Imaging (P.T.C., K.L.L.) and Division of Gastroenterology and Hepatology, Department of Internal Medicine (M.S.W., W.C.L.), National Taiwan University Hospital, National Taiwan University College of Medicine, Taipei, Taiwan; Institute of Applied Mathematical Sciences (T.W., D.C., W.W.) and Departments of Computer Science and Information Engineering (P.W.) and Internal Medicine, College of Medicine (M.S.W., W.C.L.), National Taiwan University, No. 1, Section 4, Roosevelt Rd, Taipei 10617, Taiwan; Department of Medical Imaging, National Taiwan University Cancer Center, Taipei, Taiwan (K.L.L.); NVIDIA, Bethesda, Md (H.R.R.); and National Health Insurance Administration, Ministry of Health and Welfare, Taipei, Taiwan (P.C.L.). Received March 5, 2022; revision requested April 22; revision received July 15; accepted August 1. **Address correspondence to** W.W. (email: wwang@ntu.edu.tw).

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* P.T.C. and T.W. contributed equally to this work.

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See also the editorial by Aisen and Rodrigues in this issue.

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Background: Approximately 40% of pancreatic tumors smaller than 2 cm are missed at abdominal CT.

Purpose: To develop and to validate a deep learning (DL)–based tool able to detect pancreatic cancer at CT.

Materials and Methods: Retrospectively collected contrast-enhanced CT studies in patients diagnosed with pancreatic cancer between January 2006 and July 2018 were compared with CT studies of individuals with a normal pancreas (control group) obtained between January 2004 and December 2019. An end-to-end tool comprising a segmentation convolutional neural network (CNN) and a classifier ensembling five CNNs was developed and validated in the internal test set and a nationwide real-world validation set. The sensitivities of the computer-aided detection (CAD) tool and radiologist interpretation were compared using the McNemar test.

Results: A total of 546 patients with pancreatic cancer (mean age, 65 years ± 12 [SD], 297 men) and 733 control subjects were randomly divided into training, validation, and test sets. In the internal test set, the DL tool achieved 89.9% (98 of 109; 95% CI: 82.7, 94.9) sensitivity and 95.9% (141 of 147; 95% CI: 91.3, 98.5) specificity (area under the receiver operating characteristic curve [AUC], 0.96; 95% CI: 0.94, 0.99), without a significant difference ($P = .11$) in sensitivity compared with the original radiologist report (96.1% [98 of 102]; 95% CI: 90.3, 98.9). In a test set of 1473 real-world CT studies (669 malignant, 804 control) from institutions throughout Taiwan, the DL tool distinguished between CT malignant and control studies with 89.7% (600 of 669; 95% CI: 87.1, 91.9) sensitivity and 92.8% specificity (746 of 804; 95% CI: 90.8, 94.5) (AUC, 0.95; 95% CI: 0.94, 0.96), with 74.7% (68 of 91; 95% CI: 64.5, 83.3) sensitivity for malignancies smaller than 2 cm.

Conclusion: The deep learning–based tool enabled accurate detection of pancreatic cancer on CT scans, with reasonable sensitivity for tumors smaller than 2 cm.

Online supplemental material is available for this article.

Pancreatic cancer (PC) has the lowest 5-year survival rate among cancers and is projected to become the second leading cause of cancer death in the United States by 2030 (1). Because the prognosis worsens precipitously once the tumor grows larger than 2 cm, early detection represents the most effective strategy to improve the dismal prognosis of PC (2). CT is the major imaging modality used to help detect PC, but its sensitivity for small tumors is modest, with approximately 40% of tumors smaller than 2 cm being missed (3). Furthermore, the diagnostic performance of CT is interpreter dependent and may be influenced by disparities in radiologist availability and expertise. Therefore, an effective tool to supplement radiologists in improving the sensitivity for PC detection is needed and constitutes a major unmet medical need.

Recent advances in deep learning (DL) have shown great promise in medical image analysis (4). In our previous single-center proof-of-concept study, we showed that a convolutional neural network (CNN) could accurately distinguish PC from noncancerous pancreas (5). However, in that study, segmentation of the pancreas (ie, identification of the pancreas) for analysis by the CNN was manually performed by radiologists. Segmentation of the pancreas is most challenging (6) because the pancreas borders multiple organs and structures and varies widely in shape and size, especially in patients with PC. However, a clinically applicable computer-aided detection (CAD) tool should enable segmentation and classification (ie, predicting presence or absence of PC), with minimal human annotation or labor.

Abbreviations

AUC = area under the receiver operating characteristic curve, CAD = computer-aided detection, CNN = convolutional neural network, DL = deep learning, LR = likelihood ratio, NHI = National Health Insurance, PC = pancreatic cancer

Summary

An automatic end-to-end deep learning–based detection tool could detect pancreatic cancer on CT scans in a nationwide real-world test data set with 91% accuracy, without requiring manual image labeling or preprocessing.

Key Results

- A deep learning tool for pancreatic cancer detection that was developed using contrast-enhanced CT scans obtained in 546 patients with pancreatic cancer and in 733 healthy control subjects achieved 89.9% sensitivity and 95.9% specificity in the internal test set (109 patients, 147 control subjects), which was similar to the sensitivity of radiologists (96.1%; $P = .11$).
- In a validation set comprising 1473 individual CT studies (669 patients, 804 control subjects) from institutions throughout Taiwan, the deep learning tool achieved 89.7% sensitivity and 92.8% specificity in distinguishing pancreatic cancer, with 74.7% sensitivity for pancreatic cancers smaller than 2 cm.

To fulfill this unmet clinical need, in the current study, we developed a CAD tool including a segmentation CNN to identify the pancreas on CT images and an ensemble classifier comprising five classification CNNs to predict whether the pancreas harbored PC. To ascertain its generalizability, we tested the CAD tool with the test set retrospectively derived from prospectively collected CT studies from real clinical practice throughout Taiwan. Thus, our aim was to develop and validate a DL-based tool with which to detect pancreatic cancer at CT.

Materials and Methods

This study was approved by the research ethics committee (201710050RINA, 201904116RINC). Informed consent from individual patients was waived because of the retrospective design. Figure 1 summarizes the construction of data sets (details in Appendix E1 [online]). An end-to-end workflow was implemented to analyze the CT images without manual annotation or processing (Fig 2) and consisted of image preprocessing; a segmentation CNN to segment the pancreas, including the tumor (if present); and an ensemble of five classification CNNs to determine whether the pancreas harbored PC. Computer codes are available online (*https://github.com/medalab-dladpcfncp/end_to_end_workflow*).

Local Data Set

Contrast-enhanced portal venous phase CT studies in 546 patients with PC diagnosed between January 2006 and July 2018 were extracted from the imaging archive of a tertiary referral center for further manual labeling and analysis using convenience sampling. Inclusion criteria were as follows: *(a)* age of at least 18 years; *(b)* findings recorded in cancer registry; and *(c)* pancreatic adenocarcinoma confirmed at histology or cytology. If multiple studies were identified in a patient, then only the study immediately preceding the PC diagnosis was used.

CT studies in 1465 individuals with a normal pancreas obtained between January 2004 and December 2019 were extracted from the imaging archive; these individuals comprised the control group. The control studies were selected from liver or renal donors and randomly selected individuals with a normal or unremarkable pancreas (details in Appendix E1 [online]). The patients with PC were randomly divided (8:2 ratio) into the training and validation set and the local test set, respectively. The control group was randomly divided into two portions, one for constructing the training and validation set and the local test set and the other as part of the nationwide test set. Section thickness and image size were 0.7–1.5 mm and 512 × 512 pixels, respectively. CT scans were obtained in the portal venous phase, 70–80 seconds after intravenous administration of contrast medium.

National Health Insurance Data Set

The National Health Insurance (NHI) test set was provided by the Applications of Artificial Intelligence in Medical Images Database of NHI of Taiwan. The NHI is a single-payer comprehensive compulsory health insurance system covering in- and outpatient care for 99.8% of the Taiwanese population. Among the 23 164 health care facilities throughout Taiwan, 21 463 (93%) have a contract with the NHI (7). All patients with newly confirmed PC between January 2018 and July 2019 throughout Taiwan were identified by searching the registry of the NHI Major Illness/Injury Certificate (details in Appendix E1 [online]). For selection of control subjects with a normal pancreas, we identified all kidney donors and liver donors during the same period and extracted the CT studies performed for predonation evaluation from the NHI database. Demographic information and the imaging protocol used were unavailable. The CT studies of control subjects were reviewed, and the absence of radiologic abnormalities in the pancreas was confirmed by a radiologist (P.T.C.). In total, CT studies of 669 patients with PC and 72 control subjects were extracted from the NHI image database and were further combined with CT studies of 732 control subjects from the tertiary referral center imaging archive to construct the nationwide test data set.

Training of Segmentation and Classification Models

We had previously trained a segmentation CNN (8–10) using local data with radiologist-labeled pancreas and tumor and three US data sets with pancreatic parenchyma and masses labeled (11–13). The tumor and pancreas segmented by the segmentation CNN were combined into a volume of interest to be analyzed by the classification CNNs; therefore, the performance of the classification CNNs would not be negatively affected if the tumor was segmented as pancreas. To provide more robust performance and quantitative information on the confidence of the classification, five independently operating classification CNNs were trained using the same model structure but different subsets of the training and validation set from the tertiary referral center (437 patients with PC, 586 control subjects). CT was considered to show PC if the number of positive-predicting CNNs was equal to or greater than the smallest number yielding a positive likelihood ratio (LR) greater than one in the validation set (details in Appendix E1 [online]).

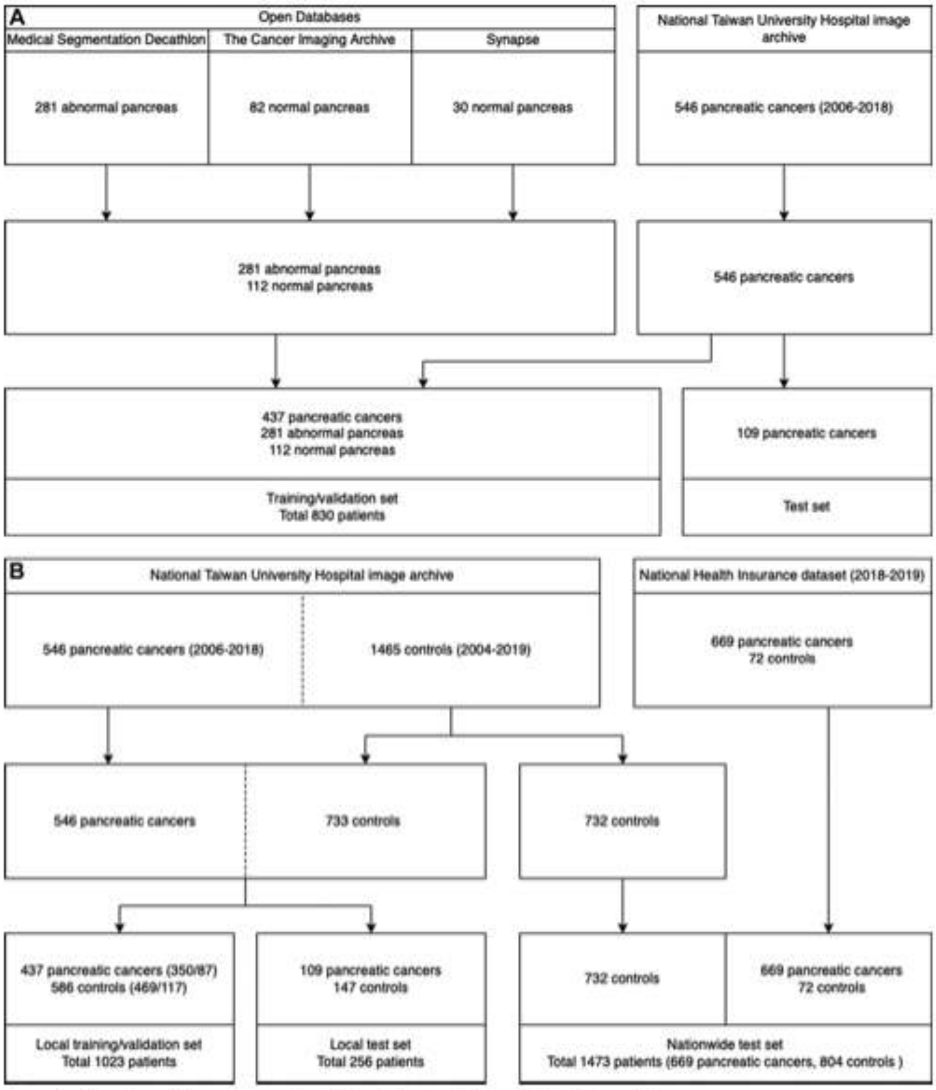


Figure 1: Data sets for the **(A)** segmentation model and **(B)** local and nationwide data sets for classification models.

Visual Assessment of Tumor Location Identified by the Segmentation Model

Consensus review by two experienced abdominal radiologists (P.T.C., K.L.L.) was conducted to assess whether the segmentation model correctly predicted the location of the tumor in the CT study classified as PC by the ensemble classifier. To explore whether the secondary signs of PC might have contributed to classification, the dilated pancreatic duct was masked by randomly selected neighboring pixels of the pancreas parenchyma.

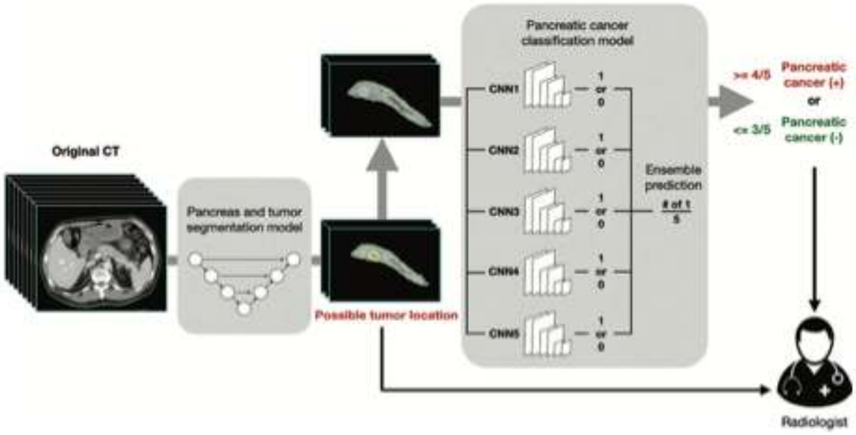


Figure 2: Workflow of the deep learning–based computer-aided detection tool. The segmentation masks passed from the segmentation convolutional neural network (CNN) to the classification CNNs included the pancreas and tumor (if present) combined, without separate delineation or identification between the pancreas and tumor. Solid arrows indicate output of the computer-aided detection tool.

Table 1: Characteristics of Pancreatic Cancer and Control Groups in Local Data Sets

Variable	Training and Validation Set		Test Set		All	
	PC Group	Control Group	PC Group	Control Group	PC Group	Control Group
No. of individuals	437	536	109	147	546	683
Age (y)	65 ± 12	54 ± 16	64 ± 12	55 ± 16	65 ± 12	54 ± 16
Sex						
Female	201 (46)	235 (44)	48 (44)	74 (50)	249 (46)	309 (49)
Male	236 (54)	301 (56)	61 (56)	73 (50)	297 (54)	374 (51)
Stage						
I	23 (5)	NA	4 (4)	NA	27 (5)	NA
II	175 (400)	NA	46 (42)	NA	221 (40)	NA
III	66 (15)	NA	13 (12)	NA	79 (14)	NA
IV	173 (40)	NA	46 (42)	NA	219 (40)	NA
Tumor size (cm)	2.9 (2.1–4.4)	NA	2.6 (2.1–3.9)	NA	2.9 (2.1–4.3)	NA

Note.—Continuous variables are presented as either means ± SDs or as medians with IQRs in parentheses. Categorical variables are presented as number of individuals, with the percentage in parentheses. Stage and tumor size are only available in patients with pancreatic cancer. NA = not applicable.

Statistical Analyses

The performance of the segmentation CNN was evaluated with Dice score per patient. The performance of each classification CNN and the ensemble classifier was assessed in various test sets to ascertain the respective sensitivity, specificity, and accuracy and the exact 95% CI for each. The sensitivity of the CAD tool was further assessed with stratification by tumor size and stage and was compared with sensitivity of radiologist interpretation as assessed in the original radiology report (details in Appendix E1 [online]). The receiver operating characteristic curves of each classification CNN and the ensemble classifier were constructed by plotting sensitivity against 1-specificity, as the threshold varied from 0 to 1. The area under the receiver operating characteristic curve (AUC) and its asymptotic 95% CI were calculated to evaluate the performance of the classifiers. LR was calculated as the ratio between the probability of the specific test result in individuals with the disease and the probability of the specific test result in those without the disease (14). The Fisher exact test and the Mann-Whitney *U* test were used to compare categorical and continuous variables between groups, respectively. The exact McNemar test was used to compare sensitivity for PC between the CAD tool and radiologist interpretation for CT studies in which radiologist reports were available (15). Trend analysis for proportions (ie, *P* trend) were implemented in STATA 17 (Stata, College Station, Tex) (16). Statistical analysis was conducted (T.W., D.C.), and $P < .05$ was considered to indicate a significant difference.

Results

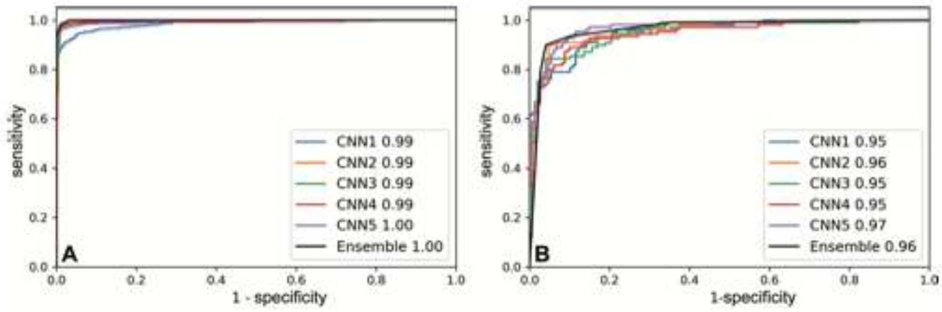
Data Set Characteristics

In the local data set, the mean age of patients with PC (54% male, 46% female) was 65 years, and the mean age of control subjects (51% male, 49% female) was 54 years (Table 1). The median age of patients with PC in Taiwan was 67 years in male patients and 69 years in female patients, with a slight male predominance (54%) (17). There were 106 healthy donors in the local training and validation set and 22 in the local test set. The nationwide test set included 210 healthy donors, 72 from the NHI and 138 from the tertiary referral center. In the local test set, the segmentation model yielded median Dice scores of 0.70 (IQR, 0.63–0.75) and 0.54 (IQR, 0.28–0.73) in segmenting the pancreas and the tumor, respectively, and 0.76 (IQR, 0.71–0.80) for the pancreas and tumor combined. The performance of the segmentation model could not be assessed in the NHI data set because segmentation of the pancreas and tumor by radiologists as the ground truth was not feasible due to NHI regulations.

Differentiation between PC and Control Groups and Associated Positive Likelihood Ratios

Figure 3A–3C shows ROC curves of classification CNNs in the training and validation set, local test set, and nationwide test set, respectively. The AUCs of the five classification CNNs ranged from 0.95 to 0.97 in the local test set and from 0.94 to 0.95 in the nationwide test set. The final classification (with PC vs without PC) was determined as 4 in the training phase because the corresponding positive LR began to exceed 1 at this threshold in the validation set (Table 2). In the local test set of 256 individuals, the ensemble classifier achieved 89.9% sensitivity (98 of 109; 95% CI: 82.7, 94.9), 95.9% specificity (141 of 147; 95% CI: 91.3, 98.5), and 93.4% accuracy (239 of 256; 95% CI: 89.6, 96.1) (AUC, 0.96; 95% CI: 0.94, 0.99) in distinguishing CT images in the PC group from those in the control group (Table 3, Fig 3B). In the nationwide test set of 1473 individuals, the ensemble classifier achieved 89.7% sensitivity (600 of 669, 95% CI: 87.1, 91.9), 92.8% specificity (746 of 804, 95% CI: 90.8, 94.5), and 91.4% accuracy (1346 of 1473; 95% CI: 89.8, 92.8) (AUC, 0.95; 95% CI: 0.94, 0.96] (Table 3, Fig 3C). There is no statistical difference (P = .20) between specificity in the 72 control patients in the NHI database (97.2%; 95% CI: 90.3, 99.7) and specificity in the 732 control patients in the local data set (92.3%; 95% CI: 90.2, 94.2).

In both the local test set and the nationwide test set, the number of classification CNNs predicting PC was positively associated with the positive LR (P_{trend} = .04 and P_{trend} = .03, respectively) (Table 2). The positive LRs for prediction as PC by four and five CNNs were 2.22 (95% CI: 1.46, 3.37) and 25.01 (95% CI: 17.10, 36.56), respectively, in the nationwide test set and 8.09 (95% CI: 1.85, 35.42) and 29.0 (95% CI: 10.98, 76.60), respectively, in the local test set.



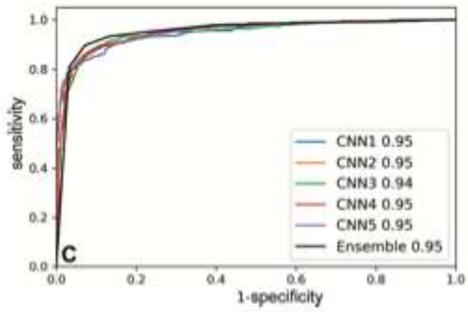


Figure 3: Receiver operating characteristic curves of the classification models in the **(A)** training and validation set, **(B)** local test set, and **(C)** nationwide test set. CNN = convolutional neural network. (Fig 3 continues).

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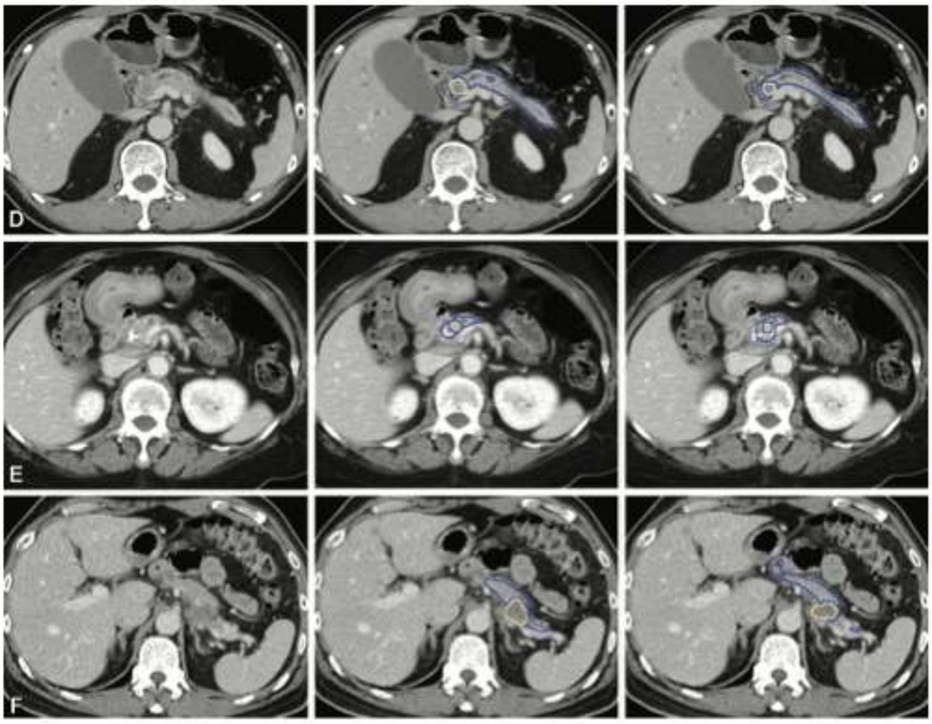

Figure 3 (continued): Representative CT scans (left column) with tumor at the pancreas **(D)** head, **(E)** body, and **(F)** tail show correspondence in tumor location between manual segmentation by radiologists (middle column) and predictions by the segmentation model (right column). Blue outline indicates the pancreas; yellow outline indicates tumor.

Table 2: Positive Likelihood Ratio according to Number of Classification Convolutional Neural Networks Predicting as Having Pancreatic Cancer

	Local Training and Validation set			Local Test Set			Nationwide Test Set–National Health Insurance Data set		
No. of CNNs	PC Group	Control Group	Positive LR	PC Group	Control Group	Positive LR	PC Group	Control Group	Positive LR
5	409	0	Infinity	86	4	29.00 (10.98, 76.60)	541	26	25.01 (17.10, 36.56)
4	19	6	4.25 (1.71, 10.54)	12	2	8.09 (1.85, 35.42)	59	32	2.22 (1.46, 3.37)
3	9	13	0.93 (0.40, 2.15)	4	9	0.60 (0.19, 1.90)	24	48	0.60 (0.37, 0.97)
2	0	14	0	2	13	0.21 (0.05, 0.90)	13	82	0.19 (0.11, 0.34)
1	0	75	0	4	24	0.22 (0.08, 0.63)	18	132	0.16 (0.10, 0.27)
0	0	478	0	1	95	0.01 (0.00, 0.10)	14	484	0.03 (0.02, 0.06)

Note.—The threshold of the number of positive-predictive convolutional neural networks (CNNs) for predicting pancreatic cancer by the ensemble classifier was 4 based on the finding that positive likelihood ratio (LR) exceeded 1 in the local validation set at this threshold. Data in parentheses are 95% CIs. PC = pancreatic cancer.

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Table 3: Performance of Computer-aided Detection Tool and Radiologists Based on the Original Report in Differentiating Between CT Studies with and Those without Pancreatic Cancer

					CAD vs Radiologist			
Test Set	Sensitivity (%)	Specificity (%)	Accuracy (%)	AUC	Sensitivity of CAD (%)	Sensitivity of Radiologist (%)	Difference	*P* Value
Local training and validation set (cancer group: 437 patients; control group: 586 patients)	97.9 (96.1, 99.1) [428/437]	99.0 (97.8, 99.6) [580/586]	98.5 (97.6, 99.2) [1008/1023]	1.00 (1.00, 1.00)	97.8 (95.9, 99.0) [406/415]*	96.1 (93.8, 97.8) [399/415]*	0.017 (−0.006, 0.040)	.17
Local test set (cancer group: 109 patients; control group: 147 patients)	89.9 (82.7, 94.9) [98/109]	95.9 (91.3, 98.5) [141/147]	93.4 (89.6, 96.1 [239/256]	0.96 (0.94, 0.99)	90.2 (82.7, 96.2) [92/102]†	96.1 (90.3, 98.9) [98/102]†	−0.059 (−0.128, 0.010)	.11
Nationwide test set (cancer group: 669 patients; control group: 804 patients)	89.7 (87.1, 91.9) [600/669]	92.8 (90.8, 94.5) [746/804]	91.4 (89.8, 92.8) [1346/1473]	0.95 (0.94, 0.96)	...	...	...	...

Note.—The ensemble classifier deemed CT findings positive if four or more of the five classification CNNs deemed it positive. Data in parentheses are 95% CIs. Data in brackets are numerators and denominators used to calculate percentages. AUC = area under the receiver operating characteristic curve, CAD = computer-aided detection.

* Excluding 22 patients without radiologist report.

† Excluding seven patients without radiologist report.

Comparison between Model Segmentation and Radiologist Interpretation with Respect to Tumor Location

In the local test set, 98 PC cases were correctly classified by the ensemble classifier, and the segmentation model-predicted tumor location was correct in 86 (87.8%) (Fig 3D–3F) and immediately adjacent to the radiologist-determined tumor in three (3.1%). In the remaining nine (9.2%) PC cases with a median tumor size of 2.3 cm, the location of the segmentation model-predicted tumor was inaccurate. In the nine patients with PC in which segmentation CNN-predicted tumor location was inaccurate, the segmentation model-predicted pancreas encompassed the tumor in six patients and included approximately half of the tumor in one patient. In the remaining two cases, the segmentation CNN classified the tumor as neither pancreas nor tumor but correctly segmented the nontumorous portion of the pancreas, which displayed secondary signs of PC (18), including upstream dilation of the pancreatic duct with abrupt duct cutoff in one case and upstream dilation of the pancreatic duct with parenchymal atrophy in the other (Fig 4A, 4B). To understand whether the secondary signs of PC in those two patients might have contributed to their classification as PC, we repeated the analysis after masking the dilated duct in the former PC case (Fig 4A), and the case was classified as noncancerous by the ensemble classifier (Fig 5). In the other case (Fig 4B), masking of secondary signs was not feasible due to lack of normal-appearing pancreas parenchyma. Among the six control patients in whom PC was incorrectly predicted by the ensemble classifier in the local test set, no apparent tumor was identified by the segmentation CNN in four cases. In the remaining two control patients, collateral veins secondary to idiopathic portal vein thrombosis were segmented as tumor in one patient by the segmentation CNN (Fig 4C); in the other, the pancreas parenchyma beside a biliary stent for palliation of obstructive jaundice from hepatocellular carcinoma was segmented as tumor (Fig 4D).

Sensitivity according to Tumor Size and Stage and Comparison with Radiologist Interpretation

In the local test set excluding seven patients without formal radiologist reports, there was no significant difference between the sensitivity of the CAD tool and that of the original radiologist report (CAD tool, 90.2%; radiologist report, 96.1%; $P = .11$) (Table 3). Four PCs were missed by radiologists, of which two (1.3 and 3.3 cm in size) were correctly classified by the ensemble classifier. Stratified analysis according to primary tumor size and stage also showed similar sensitivity between the CAD tool and attending radiologists (Table 4). For tumors smaller than 2 cm, the sensitivities for detecting PC with the CAD were 87.5% (21 of 24, 95% CI: 67.6, 97.3) in the local test set and 74.7% (68 of 91, 95% CI: 64.5, 83.3) in the nationwide test set. Radiologist reports and information on cancer stage were not available in the NHI data set. Comparison between the CAD tool and radiologists with respect to specificity was not feasible, given that control patients were selected based on the statement of a normal or unremarkable pancreas in the radiologist report, because otherwise it was inappropriate to exclude the possibility of PC.

Discussion

Enhancing the sensitivity of CT for small or early pancreatic cancer (PC) is a major unmet clinical need. This study developed and validated an end-to-end deep learning (DL)–based computer-aided detection (CAD) tool to differentiate between CT studies with and those without PC without requiring manual image labeling or preprocessing. In a test set retrospectively derived from prospectively collected real-world images from institutions throughout Taiwan, the tool achieved 89.7% sensitivity (74.7% for PCs <2 cm) and 92.8% specificity, demonstrating high robustness and generalizability. No significant difference in sensitivity was noted between the CAD tool (90.2%) and attending radiologists (96.1%) in a tertiary academic institution with a large volume of patients with PC (P = .11); therefore, the CAD tool might have higher sensitivity compared with radiologists who are less experienced with PC.

Generalizability to new data sets is a prerequisite for clinical applications, but the performance of CNNs often deteriorates significantly in external validation (19–22), and few studies of medical image analysis by artificial intelligence undergo external validation (23). A recent study also investigated end-to-end diagnosis of PC by DL and reported an accuracy of 87.6% (24), but no external validation was attempted.

Our current study had several methodologic advances compared with the previous study (5). To obviate the need for manually segmenting the pancreas, we trained a segmentation CNN based on a coarse-to-fine neural architecture search, which is most suitable for pancreas segmentation (8,9). The segmentation CNN was trained with images from multiple institutions and races or ethnicities and yielded Dice scores comparable to the best-performing pancreas segmentation DL models reported (25–27). Second, we adopted three-dimensional volumetric analyses in segmentation and classification, whereas the previous study used two-dimensional analyses,

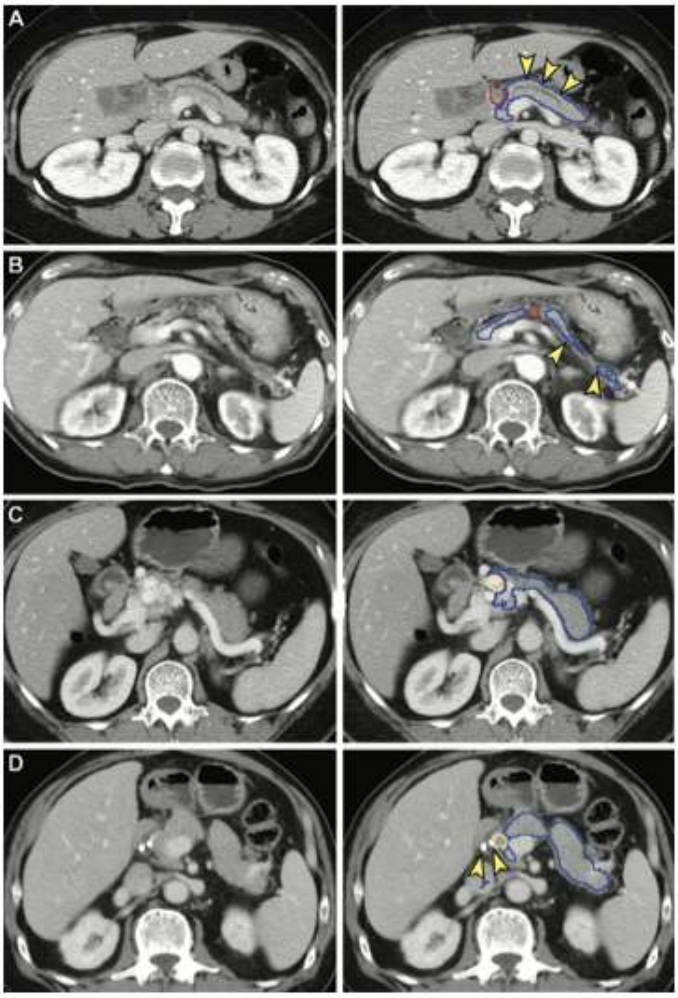

Figure 4: False-negative **(A, B)** and false-positive **(C, D)** tumor segmentation by the segmentation model. Blue and yellow outlines indicate normal pancreas and tumor segmented with the segmentation model, respectively. Images in the left column are original unannotated CT scans. **(A, B)** Tumors (red outline) were not segmented by the segmentation convolutional neural network. The upstream pancreas shows secondary signs of pancreatic cancer, including dilation of the pancreatic duct with abrupt cutoff (arrowhead in **A**) and parenchymal atrophy with dilation of the pancreatic duct (arrowhead in **B**). **(C)** Collateral veins secondary to idiopathic portal vein thrombosis were incorrectly segmented as tumor by the segmentation model. **(D)** Pancreatic parenchyma adjacent to biliary stents (arrowhead) placed for relieving obstructive jaundice from hepatocellular carcinoma was incorrectly segmented as tumor by the segmentation model.

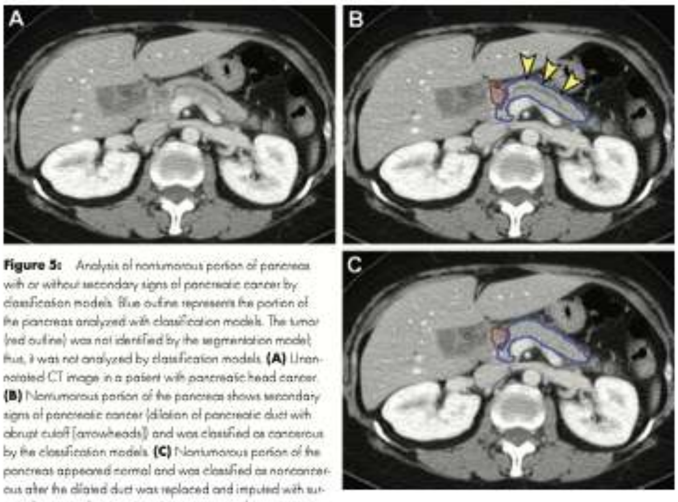

Figure 5: Analysis of nontumorous portion of pancreas with or without secondary signs of pancreatic cancer by classification models. Blue outline represents the portion of the pancreas analyzed with classification models. The tumor (red outline) was not identified by the segmentation model; thus, it was not analyzed by classification models. **(A)** Unannotated CT image in a patient with pancreatic head cancer. **(B)** Nontumorous portion of the pancreas shows secondary signs of pancreatic cancer (dilation of pancreatic duct with abrupt cutoff [arrowheads]) and was classified as cancerous by the classification models. **(C)** Nontumorous portion of the pancreas appeared normal and was classified as noncancerous after the dilated duct was replaced and imputed with surrounding normal-appearing pancreas parenchyma.

losing information between neighboring CT sections and ignoring the spatial correlations among sections. Integrating the segmentation and classification CNNs enabled automatic analysis of a CT study in approximately 30 seconds, supporting the feasibility for clinical deployment.

This CAD tool could provide a multitude of information to assist clinicians. Besides determining whether the images showed PC, the tool could indicate the region of suspicion to expedite radiologist interpretation. In approximately 90% of PCs correctly classified by the CAD tool, the segmentation CNN correctly identified the tumor location. As the segmentation CNN might falsely segment normal pancreas parenchyma as tumor when encountering uncommon conditions or artifacts, the area determined as tumor by the segmentation CNN should be interpreted as the probable location of the tumor only in cases classified as PC by the classification CNNs rather than being overinterpreted as another set of classification. Furthermore, the CAD tool could provide the positive LR, a measure of the confidence of the classification (PC vs non-PC), which could be multiplied with the pretest odds determined based on clinical parameters and experience to derive the posttest odds and probability of PC, thereby better informing the subsequent diagnostic-therapeutic process than a simple binary classification.

The comparable sensitivity between the CAD tool and experienced radiologists at a tertiary referral center supports the idea that the CAD tool might be useful for reducing the miss rate attributed to disparities in expertise. The radiologists' high sensitivity for PCs smaller than 2 cm observed in this tertiary referral center might not be generalizable to other institutions. Previous research showed that approximately 40% of PCs smaller than 2 cm were missed on CT scans (28), whereas our CAD tool achieved 87.5% and 74.7% sensitivity for PCs smaller than 2 cm in the local and nationwide test sets, respectively.

The interesting finding that the classification CNNs correctly classified two cases of PC by analyzing only the nontumorous portion of the pancreas underscored the diagnostic value of secondary signs of PC, including pancreatic duct dilatation, upstream pancreatic parenchymal atrophy, and abrupt cutoff of the pancreatic duct (18). Besides features in the tumor, secondary signs in the nontumorous portion of the pancreas are important clues to occult PCs and thus should be taken advantage of in developing CAD tools for PC. Our explorative analysis showed that the nontumorous portion of the pancreas with secondary signs of PC and the portion without such signs were classified as with PC and without PC, respectively. Our results suggest that the classification CNNs might have learned the secondary signs of PC, in line with the notion that DL can spontaneously capture distinguishing imaging features through learning from examples (29,30).

This study had limitations. First, radiologist reports were unavailable in the NHI data set; thus, comparison between the CAD tool and radiologists was not feasible. Second, the difference in sensitivity between the CAD tool and radiologists at a tertiary referral center might not be generalizable to other institutions. Third, although the nationwide NHI data set included variations in imaging parameters and quality occurring in real clinical practice and thus represented a most rigorous test set, the Taiwanese population in the test set was predominantly Asian and relatively homogenous in race and ethnicity. Fourth, control subjects from the NHI database were healthy donors and were likely younger. However, no significant difference in specificity was noted between control subjects from the NHI and those from local databases; thus, this limitation should not have favorably biased the results. Last, the control group did not include patients with pancreatic abnormalities other than PC, many of which require tissue sampling for confirmatory diagnosis. We seek to include other pancreatic abnormalities and prospectively assess the potential usefulness of the CAD tool in clinical settings in a future study.

In conclusion, this study developed an end-to-end deep learning–based computer-aided detection (CAD) tool that could accurately and robustly detect pancreatic cancers (PCs) on contrast-enhanced CT scans. The CAD tool may be a useful supplement for radiologists to enhance detection of PC. Our results also suggest that the classification convolutional neural networks might have learned the secondary signs of PC, which warrants further investigation. While the results of this study provide strong

Table 4: Sensitivity of Computer-sided Detection Tool and Radiologists Stratified by Tumor Stage and Size

Stage and Size	CAD Tool	Local Test Set: CAD vs Radiologist				Nationwide Test Set: CAD Tool
		Sensitivity of CAD (%)	Sensitivity of Radiologist (%)	Difference	*P* Value	
Stage						
I	50.0 (6.8, 93.2) [2/4]	50.0 (6.8, 93.2) [2/4]	75.0 (19.4, 99.4) [3/4]	−0.250 (−0.898, 0.398)	>.99	NA*
II	89.1 (76.4, 96.4) [41/46]	88.6 (75.4, 96.3) [39/44]†	97.7 (88.0, 99.9) [43/44]†	−0.091 (−0.195, 0.013)	.13	NA*
III	92.3 (64.0, 99.8) [12/13]	91.7 (61.5, 98.8) [11/12]‡	91.7) (61.5, 98.8) [11/12]‡	0.000 (−0.221, 0.221)	>.99	NA*
IV	93.5 (82.1, 98.6) [43/46]	95.2 (83.8, 99.4) [40/42]§	97.6 (87.4, 99.9) [41/42]§	−0.024 (−0.104, 0.056)	>.99	NA*
Size (cm)						
<2	87.5 (67.6, 97.3) [21/24]	86.4 (65.1, 97.1) [19/22]‖	95.5 (77.2, 99.9) [21/22]‖	−0.091 (−0.258, 0.076)	.63	74.7 (64.5, 83.3) [68/91]
2–4	89.8 (79.2, 96.2) [53/59]	90.9 (80.0, 97.9) [50/55]#	96.4) (87.5, 99.6) [53/55]#	−0.055 (−0.146, 0.036)	.38	91.4 (88.0, 94.0) [339/371]
>4	92.3 (74.9, 99.1) [24/26]	92.0 (74.0, 99.0) [23/25]**	96.0 (79.6, 99.9) [24/25]**	−0.040 (−0.171, 0.091)	>.99	93.2 (88.9, 96.3) [193/207]

Note.—Data in parentheses are 95% CIs, and data in brackets are numerators and denominators used to calculate percentages. The cancer stage in the local test set was extracted from the cancer registry and determined in accordance with current standards according to the American Joint Committee on Cancer (seventh edition) based on histologic findings in patients undergoing surgical resection and on complete staging work-up in patients not undergoing surgery. Tumor size was determined according to the histologic report if the patient underwent surgery. Otherwise, tumor size was based on the CT report. In the nationwide data set, information on cancer stage was not available, and tumor size was measured on CT images and was recorded by a radiologist. CAD = computer-aided detection, NA = not applicable.

* Information on tumor stage was not available in the nationwide test set.

† Excluding two patients without radiologist report.

‡ Excluding one patient without radiologist report.

§ Excluding four patients without radiologist report.

‖ Excluding two patients without radiologist report.

\# Excluding four patients without radiologist report.

** Excluding one patient without radiologist report.

support for the generalizability of the CAD tool in the Taiwanese and perhaps Asian populations, the performance of the CAD tool in other populations needs to be evaluated further.

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References

1. Rahib L, Smith BD, Aizenberg R, Rosenzweig AB, Fleshman JM, Matrisian LM. Projecting cancer incidence and deaths to 2030: the unexpected burden of thyroid, liver, and pancreas cancers in the United States. Cancer Res 2014;74(11):2913–2921.
2. Chari ST. Detecting early pancreatic cancer: problems and prospects. Semin Oncol 2007;34(4):284–294.
3. Kang JD, Clarke SE, Costa AF. Factors associated with missed and misinterpreted cases of pancreatic ductal adenocarcinoma. Eur Radiol 2021;31(4):2422–2432.
4. Chen PT, Chang D, Wu T, Wu MS, Wang W, Liao WC. Applications of artificial intelligence in pancreatic and biliary diseases. J Gastroenterol Hepatol 2021;36(2):286–294.
5. Liu KL, Wu T, Chen PT, et al. Deep learning to distinguish pancreatic cancer tissue from non-cancerous pancreatic tissue: a retrospective study with cross-racial external validation. Lancet Digit Health 2020;2(6):e303–e313.
6. Yao X, Song Y, Liu Z. Advances on pancreas segmentation: a review. Multimed Tools Appl 2020;79(9):6799–6821.
7. National Health Insurance Administration Ministry of Health and Welfare. https://www.nhi.gov.tw/English/Content_List.aspx?n=8FC0974BBFEFA56D&topn=ED4A30E51A609E49. Published 2016. Accessed May 21, 2021.
8. Wang P, Shen C, Roth HR, et al. Automated Pancreas Segmentation Using Multi-institutional Collaborative Deep Learning. In: Domain Adaptation and Representation Transfer, and Distributed and Collaborative Learning. Cham, Switzerland: Springer International Publishing, 2020; 192–200.
9. Shen C, Wang P, Roth HR, et al. Multi-task federate learning for heterogeneous pancreas segmentation. In: Clinical Image-Based Procedures, Distributed and Collaborative Learning, Artificial Intelligence for Combating COVID-19 and Secure and Privacy-Preserving Machine Learning. Cham, Switzerland: Springer International Publishing, 2021; 101–110.
10. Yu Q, Yang D, Roth H, et al. C2FNAS: Coarse-to-Fine Neural Architecture Search for 3D Medical Image Segmentation. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, June 13–19, 2020. Piscataway, NJ: IEEE, 2020; 4125–4134.
11. Simpson AL, Antonelli M, Bakas S, et al. A large annotated medical image dataset for the development and evaluation of segmentation algorithms. arXiv [preprint] https://arxiv.org/abs/1902.09063. Posted February 25, 2019. Accessed February 27, 2022.
12. Pancreas-CT - The Cancer Imaging Archive (TCIA) Public Access - Cancer Imaging Archive Wiki. https://wiki.cancerimagingarchive.net/display/Public/Pancreas-CT. Accessed June 14, 2019.
13. Gibson E, Giganti F, Hu Y, et al. Multi-organ Abdominal CT Reference Standard Segmentations. . Published February 22, 2018. Accessed February 27, 2022.
14. Deeks JJ, Altman DG. Diagnostic tests 4: likelihood ratios. BMJ 2004;329(7458):168–169.

15. Zhou XH. Statistical methods in diagnostic medicine. Hoboken, NJ: Wiley, 2011.
16. Royston P. PTREND: Stata module for trend analysis for proportions. Boston College Department of Economics. https://EconPapers.repec.org/RePEc:boc:bocode:s426101. Published 2014. Accessed May 21, 2022.
17. Health Promotion Administration. https://www.hpa.gov.tw/Pages/Detail.aspx?nodeid=269&pid=14913. Published 2016. Accessed May 21, 2022.
18. Tamada T, Ito K, Kanomata N, et al. Pancreatic adenocarcinomas without secondary signs on multiphasic multidetector CT: association with clinical and histopathologic features. Eur Radiol 2016;26(3):646–655.
19. Bai W, Sinclair M, Tarroni G, et al. Automated cardiovascular magnetic resonance image analysis with fully convolutional networks. J Cardiovasc Magn Reson 2018;20(1):65.
20. Barish M, Bolourani S, Lau LF, Shah S, Zanos TP. External validation demonstrates limited clinical utility of the interpretable mortality prediction model for patients with COVID-19. Nat Mach Intell 2020;3(1):25–27.
21. Gulshan V, Peng L, Coram M, et al. Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs. JAMA 2016;316(22):2402–2410.
22. Beede E, Baylor E, Hersch F, et al. A Human-Centered Evaluation of a Deep Learning System Deployed in Clinics for the Detection of Diabetic Retinopathy. In: CHI '20: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. New York, NY: Association for Computing Machinery, 2020; 1–12.
23. Kim DW, Jang HY, Kim KW, Shin Y, Park SH. Design Characteristics of Studies Reporting the Performance of Artificial Intelligence Algorithms for Diagnostic Analysis of Medical Images: Results from Recently Published Papers. Korean J Radiol 2019;20(3):405–410.
24. Si K, Xue Y, Yu X, et al. Fully end-to-end deep-learning-based diagnosis of pancreatic tumors. Theranostics 2021;11(4):1982–1990.
25. Turečková A, Turecek T, Komínková Oplatková Z, Rodríguez-Sánchez A. Improving CT Image Tumor Segmentation Through Deep Supervision and Attentional Gates. Front Robot AI 2020;7:106.
26. Zhang Y, Wu J, Liu Y, et al. A deep learning framework for pancreas segmentation with multi-atlas registration and 3D level-set. Med Image Anal 2021;68:101884.
27. Antonelli M, Reinke A, Bakas S, et al. The Medical Segmentation Decathlon. arXiv [preprint] https://arxiv.org/abs/2106.05735. Posted June 10, 2021. Accessed February 27, 2022.
28. Dewitt J, Devereaux BM, Lehman GA, Sherman S, Imperiale TF. Comparison of endoscopic ultrasound and computed tomography for the preoperative evaluation of pancreatic cancer: a systematic review. Clin Gastroenterol Hepatol 2006;4(6):717–725; quiz 664.
29. Litjens G, Kooi T, Bejnordi BE, et al. A survey on deep learning in medical image analysis. Med Image Anal 2017;42:60–88.
30. Esteva A, Robicquet A, Ramsundar B, et al. A guide to deep learning in healthcare. Nat Med 2019;25(1):24–29.

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